

Electronics for Embedded Systems

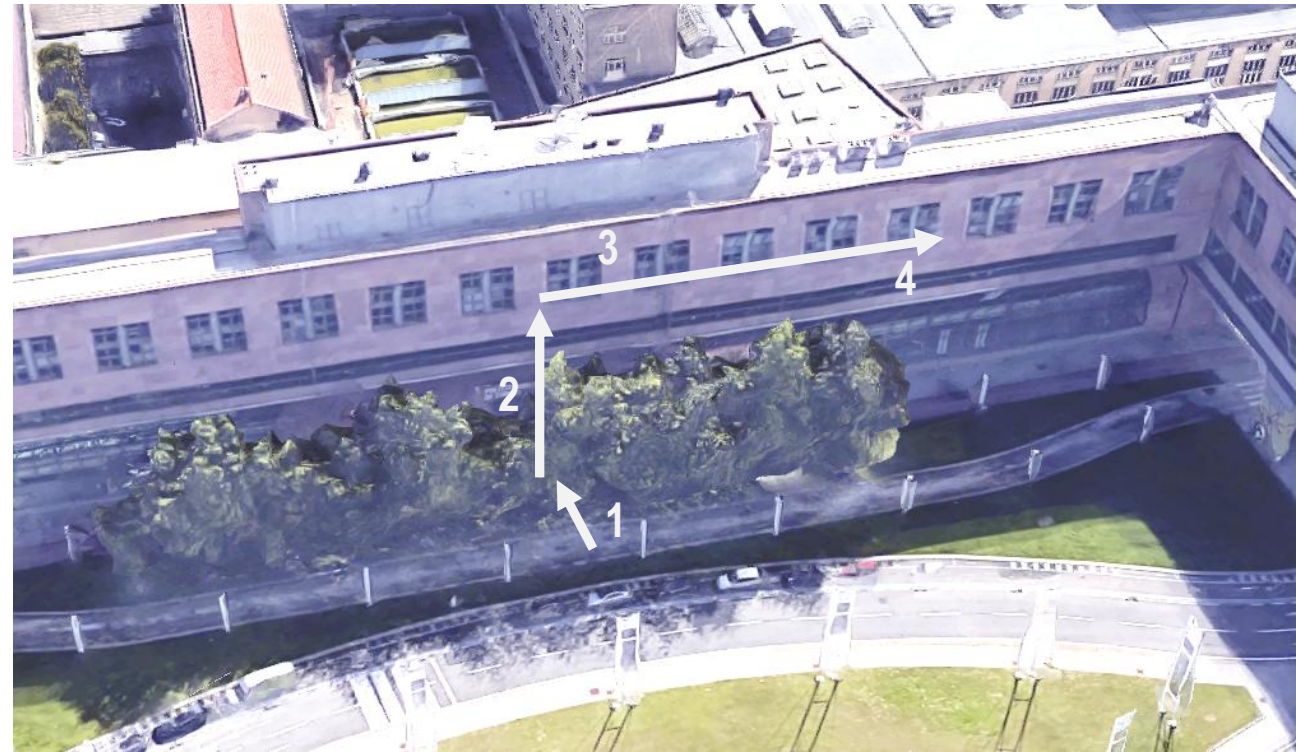
Introduction and Review

Prof. Claudio Passerone

Course presentation

- Prof. Claudio Passerone
 - Dept. of Electronics and Telecommunications (DET)
 - Third floor, Cittadella Politecnica
 1. Corso Castelfidardo 39
 2. Take elevator to third floor
 3. DET door 1, call me with the blue phone at the number 4039 (but the phone is often broken...)
 4. 7th door on the right
- email: claudio.passerone@polito.it
- phone: +39 011 0904039

Last year Telegram group
https://t.me/efes_polito



Course presentation – Timetable

• Classes

- Wednesday 8:30 – 11:30 (Room 27b)
- Thursday 11:30 – 13:00 (Room 1I)
- Friday 14:30 – 16:00 (Room 2P)

• Lab

- Five lab experiences during the course
- Two possible sessions:
 - Monday 13:00 – 16:00 (LED 3)
 - Tuesday 11:30 – 14:30 (LED 3)
- Groups of maximum 4 students
- LED, 2nd floor, Cittadella Politecnica

	Mon	Tue	Wed	Thu	Fri	Sat	Sun	
	21	22	23	24	25	26	27	Sep
	28	29	30	1	2	3	4	
	5	6	7	8	9	10	11	
	12	13	14	15	16	17	18	Oct
	19	20	21	22	23	24	25	
Lab 1 →	26	27	28	29	30	31	1	
	2	3	4	5	6	7	8	
Lab 2 →	9	10	11	12	13	14	15	Nov
	16	17	18	19	20	21	22	
Lab 3 →	23	24	25	26	27	28	29	
	30	1	2	3	4	5	6	
	7	8	9	10	11	12	13	
Lab 4 →	14	15	16	17	18	19	20	Dec
	21	22	23	24	25	26	27	
	28	29	30	31	1	2	3	
	4	5	6	7	8	9	10	Jan
Lab 5 →	11	12	13	14	15	16	17	

Course presentation – Timetable

- Weeks with only classes

	Monday	Tuesday	Wednesday	Thursday	Friday
8:30 – 10:00			Room 27b		
10:00 – 11:30					
11:30 – 13:00				Room 1I	
13:00 – 14:30					
14:30 – 16:00					Room 2P
16:00 – 17:30					
17:30 – 19:00					

- Weeks with laboratories

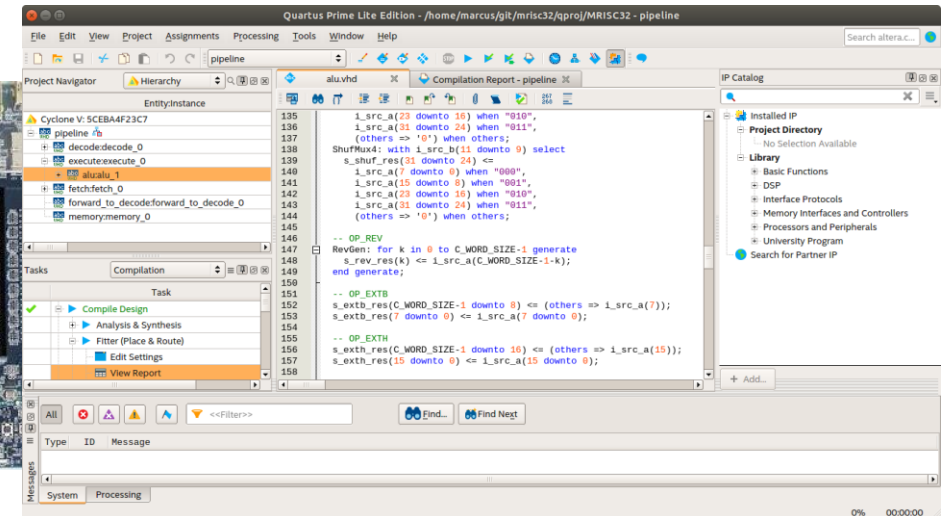
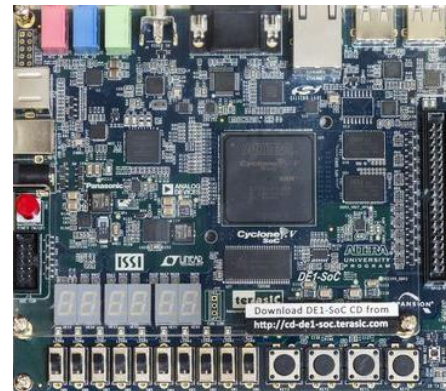
	Monday	Tuesday	Wednesday	Thursday	Friday
8:30 – 10:00			Room 3P		
10:00 – 11:30					
11:30 – 13:00		LED 3		Room 10A	
13:00 – 14:30	LED 3				
14:30 – 16:00					Room 2P
16:00 – 17:30					
17:30 – 19:00					

Course presentation – Topics

- Introduction to Embedded Systems and review of analog and digital electronics
- Memories
- Programmable Logic Devices (LAB)
- Interconnections protocols and I/O (LAB)
- Embedded processors and peripherals (LAB)
- DA and AD conversion (LAB)
- Power Management (LAB)
- Conclusions and review

Course presentation – Laboratories

- Five laboratory sessions
 - PLD, Interconnections, Processor Peripherals, AD Conversion, Power Management
 - Intel/Altera DE1-SoC board, Quartus Prime 18.1
- Groups of maximum 4 students
 - You form the groups
 - Choose Monday or Tuesday session
- Laboratory reports
 - Up to 2 bonus points at the exam
 - The bonus points never expire
 - Not compulsory
 - But you don't get bonus points if you don't submit the reports
 - I take the presence in the laboratory
 - No more than 2 absences



Course presentation – Text books

- Passerone C., *Analog and Digital Electronics for Embedded Systems*, CLUT, 2015
- A generic textbook on electronics
 - Sedra A. and Smith K., *Microelectronics Circuits*, Oxford University Press
 - Jaeger R. and Blalok T., *Microelectronic Circuit Design*, McGraw Hill
 - Millman J. and Grabel A., *Microelectronics*, McGraw Hill
 - Rabaey J., Chandrakasan A. and Nikolic B., *Digital Integrated Circuits: A Design Perspective*, Prentice Hall
- Passerone C., *Introduction to the Verilog language*, Amazon, April 2015,
<https://amzn.eu/d/gvA7CD1>

Course presentation – Additional material

- Material published on the student portal
 - Video recording of lessons
 - Virtual Classrooms of academic year 2025/26
 - Recordings of previous academic years
 - Slides
 - Datasheets
 - Past exams with solutions
 - Very detailed
 - Questions are often repeated...
 - Laboratory instructions

Course presentation – Exam

- Written exam (on paper)
 - 3 hours
 - No material
 - Both exercises and theoretical questions
 - You can reach “30 with honors” with the written exam alone
 - You can reject the final grade

Course presentation – Exam

- Optional oral exam (after the written exam)
 - Can take oral if written exam is ≥ 15
 - Mostly theoretical questions
 - Oral can increase or decrease the final grade
 - Usually 2 questions, around 45 minutes
- LAB grades (0, +0.5, +1, +1.5, +2)
 - Lab reports are not compulsory

Course presentation – Exam

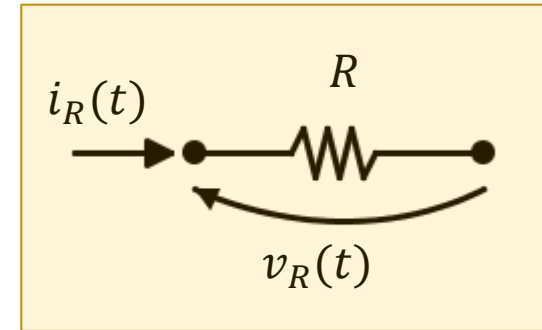
- Project
 - Individual (no groups)
 - Proposed by the student
 - There are no deadlines to make a proposal or to complete the work
 - Should cover all the topics of the course in a non-trivial way
 - Memories (not enough to just use a memory on a board)
 - Programmable Logic Devices (write some VHDL or Verilog code)
 - Interconnections (use some communication protocol)
 - Processor peripherals (program a peripheral of a processor without using a driver)
 - AD and DA conversion (not enough to just use a converter on a board)
 - Power management (power supply or actuator driving)
 - Send a description of what you want to do
 - Complete the work
 - Write a report and give me a live presentation of the project

Review

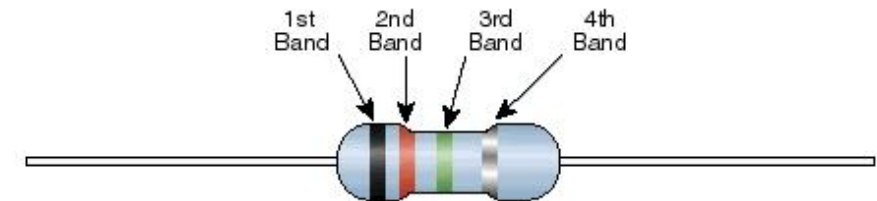
- Electrical components
 - Passive components (R, C, L)
 - Active components (diode, transistors)
 - Operational amplifiers
 - Amplifier circuits, comparators
- Digital electronics
 - Logic gates
 - Digital functions
 - Finite State Machines
 - VHDL

Resistor and resistance

- Measured in Ohms (symbol Ω)
- Ohm law
 - $v_R(t) = R i_R(t)$
 - $V_R(s) = R I_R(s)$
- Conductance
 - $g = \frac{1}{R}$
- Behaviour does not depend on frequency
- How to read the value of a resistance
 - Useful during the lab sessions



Standard EIA Color Code Table 4 Band: $\pm 2\%$, $\pm 5\%$, and $\pm 10\%$

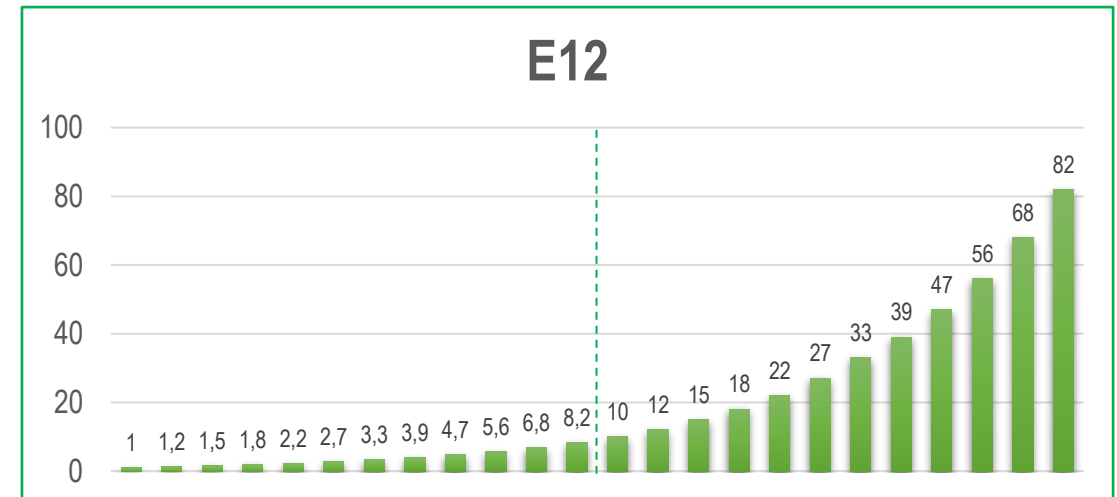


Color	1st Band (1st figure)	2nd Band (2nd figure)	3rd Band (multiplier)	4th Band (tolerance)
Black	0	0	10^0	
Brown	1	1	10^1	
Red	2	2	10^2	$\pm 2\%$
Orange	3	3	10^3	
Yellow	4	4	10^4	
Green	5	5	10^5	
Blue	6	6	10^6	
Violet	7	7	10^7	
Gray	8	8	10^8	
White	9	9	10^9	
Gold			10^{-1}	$\pm 5\%$
Silver			10^{-2}	$\pm 10\%$

Chart Provided By XICON

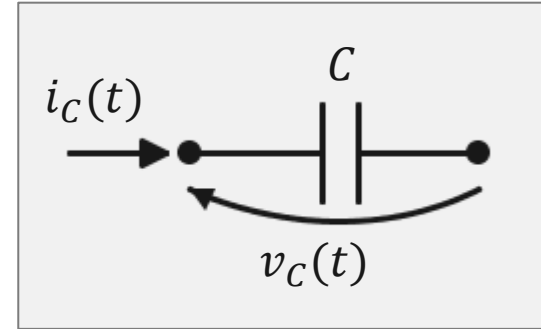
Resistor and resistance

- Standard values in E series
 - E6 (tolerance 20%)
 - 1.0, 1.5, 2.2, 3.3, 4.7, 6.8
 - E12 (tolerance 10%)
 - 1.0, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2
 - E24 (tolerance 5%)
 - 1.0, 1.1, 1.2, 1.3, 1.5, 1.6, 1.8, 2.0, 2.2, 2.4, 2.7, 3.0, 3.3, 3.6, 3.9, 4.3, 4.7, 5.1, 5.6, 6.2, 6.8, 7.5, 8.2, 9.1
 - Examples from E12 series
 - 2.2 k Ω , 2.2 M Ω , 220 k Ω , 2.2 Ω
 - 56 k Ω , 0.56 Ω , 560 Ω
 - 3.3 G Ω , 33 m Ω
 - 100 m Ω , 1 Ω , 10 Ω , 100 Ω , 1 k Ω , 10 k Ω , 100 k Ω



Capacitor and capacitance

- Measured in Farad (symbol F)
 - $i_C(t) = C \frac{dv_C(t)}{dt}$
 - $v_C(t) = \frac{1}{C} \int i_C(t) dt$
 - $V_C(s) = \frac{I_C(s)}{sC}$
- Behaviour depends on frequency
 - When $f \rightarrow 0$ it behaves as an open circuit
 - When $f \rightarrow \infty$ it behaves as a short circuit



1 mF

100 nF
± 20%

100 nF
± 10%

2.2 nF

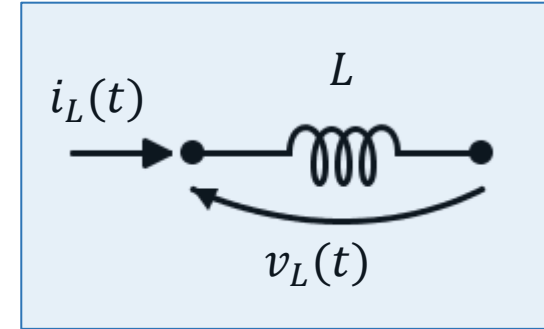
10 µF

Capacitor and capacitance

- How to read the value of a capacitor
 - Two digits: it is the value in pF
 - $33 \rightarrow 33 \text{ pF}$
 - Three digits: the first two are the value in pF, and the third is the multiplier
 - $332 \rightarrow 33 \times 10^2 \text{ pF} = 3300 \text{ pF} = 3.3 \text{ nF}$
 - If the third digit is 8 the multiplier is 10^{-2} , if it is 9 the multiplier is 10^{-1}
 - A digit, a character and a digit: the two digits are the one before and after the decimal point, the character is the unit of measure
 - $1\text{n}2 \rightarrow 1.2 \text{ nF}$
 - A character follows the last digit
 - It indicates the tolerance
 - $J = \pm 5\%$, $K = \pm 10\%$, $M = \pm 20\%$

Inductor and inductance

- Measured in Henry (symbol H)
 - $v_L(t) = L \frac{di_L}{dt}$
 - $i_L(t) = \frac{1}{L} \int v_L(t) dt$
 - $V_L(s) = sLI_L(s)$
- Behaviour depends on frequency
 - When $f \rightarrow 0$ it behaves as a short circuit
 - When $f \rightarrow \infty$ it behaves as an open circuit



Voltage and current dividers

- Voltage divider with two resistances

- $v_{out}(t) = \frac{R_2}{R_1 + R_2} v_{in}(t) = \frac{g_1}{g_1 + g_2} v_{in}(t)$

- $v_{R_1}(t) = \frac{R_1}{R_1 + R_2} v_{in}(t) = \frac{g_2}{g_1 + g_2} v_{in}(t)$

- Voltage divider with three resistances

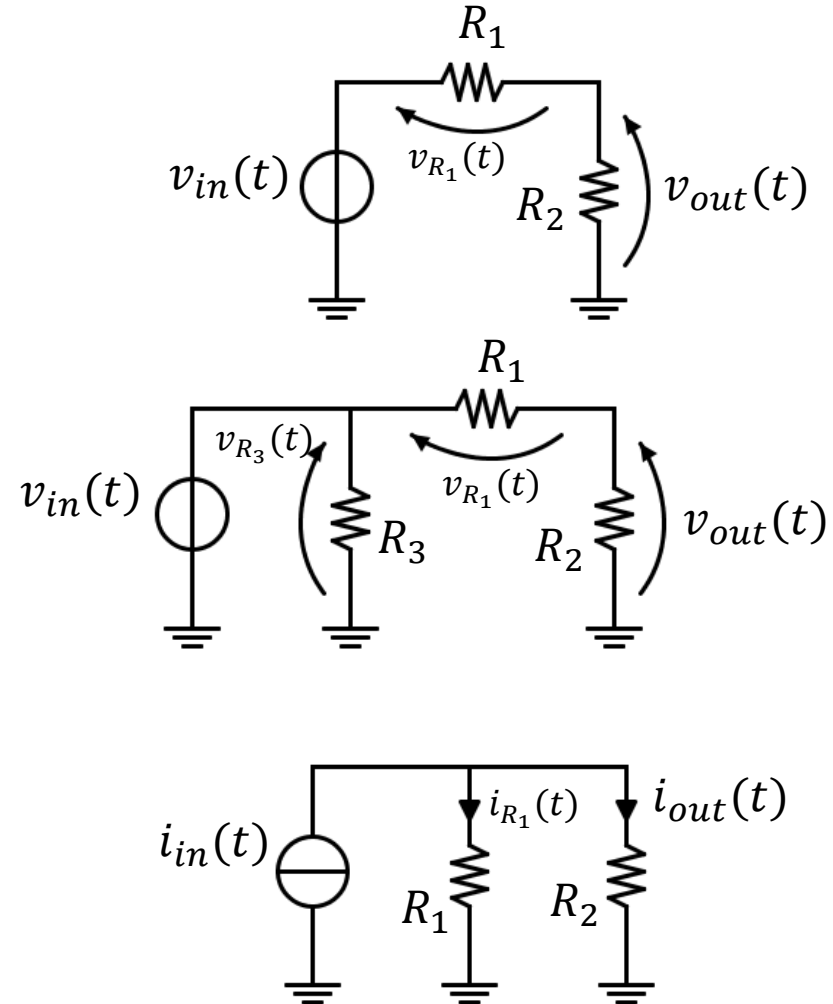
- $v_{out}(t) = \frac{R_2}{R_1 + R_2} v_{in}(t) = \frac{g_1}{g_1 + g_2} v_{in}(t)$

- $v_{R_3}(t) = v_{in}(t)$

- Current divider

- $i_{out}(t) = \frac{R_1}{R_1 + R_2} i_{in}(t) = \frac{g_2}{g_1 + g_2} i_{in}(t)$

- $i_{R_1}(t) = \frac{R_2}{R_1 + R_2} i_{in}(t) = \frac{g_1}{g_1 + g_2} i_{in}(t)$



Simple low pass filter

- Low pass filter with a resistance and a capacitance

- Kirchoff voltage law

$$v_{in}(t) - RC \frac{dv_{out}(t)}{dt} - v_{out}(t) = 0 \quad \text{with } i_C(t) = C \frac{dv_{out}(t)}{dt}$$

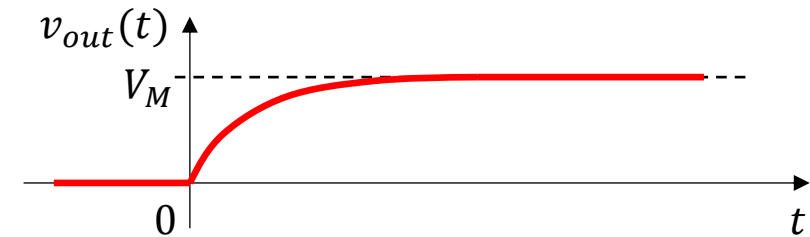
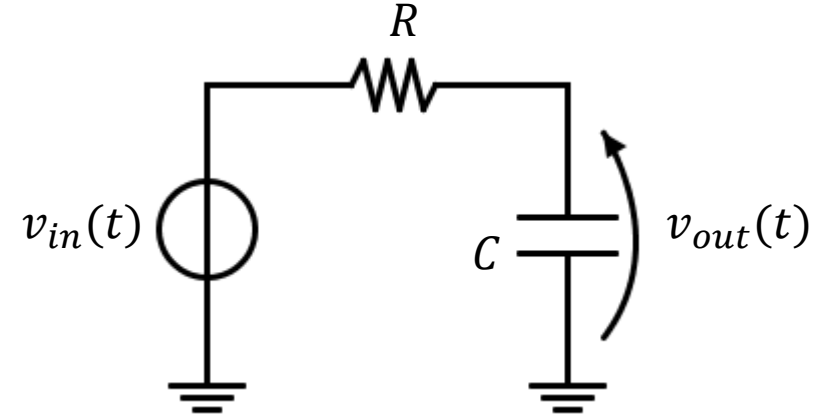
- First order linear differential equation

$$\frac{dv_{out}(t)}{dt} + \frac{1}{\tau} v_{out}(t) = \frac{1}{\tau} v_{in}(t) \quad \text{with } \tau = RC$$

$$v_{out}(t) = e^{-\frac{t}{\tau}} \left[k + \int \left(\frac{v_{in}(t)}{\tau} e^{\frac{t}{\tau}} \right) dt \right]$$

- With $v_{in}(t)$ a step of amplitude V_M

$$v_{out}(t) = V_M (1 - e^{-\frac{t}{\tau}}) \quad \text{with } t \geq 0$$



Simple low pass filter

- Low pass filter with a resistance and a capacitance

- $V_{out}(s) = \frac{1/sC}{1/sC+R} V_{in}(s) = \frac{1}{sRC+1} V_{in}(s)$

- $\frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{sRC+1}$

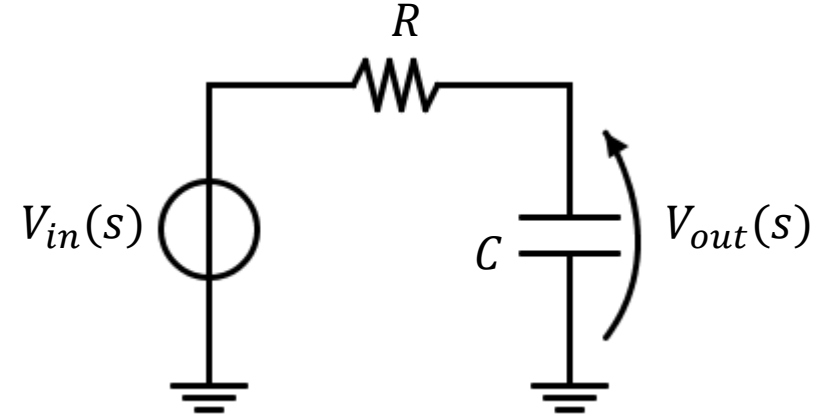
- $V_{out}(j\omega) = \frac{1}{j\omega RC+1} V_{in}(j\omega)$

- $\left| \frac{V_{out}(j\omega)}{V_{in}(j\omega)} \right| = \frac{1}{\sqrt{\omega^2 R^2 C^2 + 1}}$

- A pole is a zero of the denominator

- $s_p = -\frac{1}{RC}$

- $\omega_p = \frac{1}{RC}$



Simple low pass filter time domain response

- If V_{in} is a square wave, compute the response in the time domain

- Each fragment in the response is an exponential

- $v_{out}(t) = A + Be^{-\frac{t}{\tau}}$

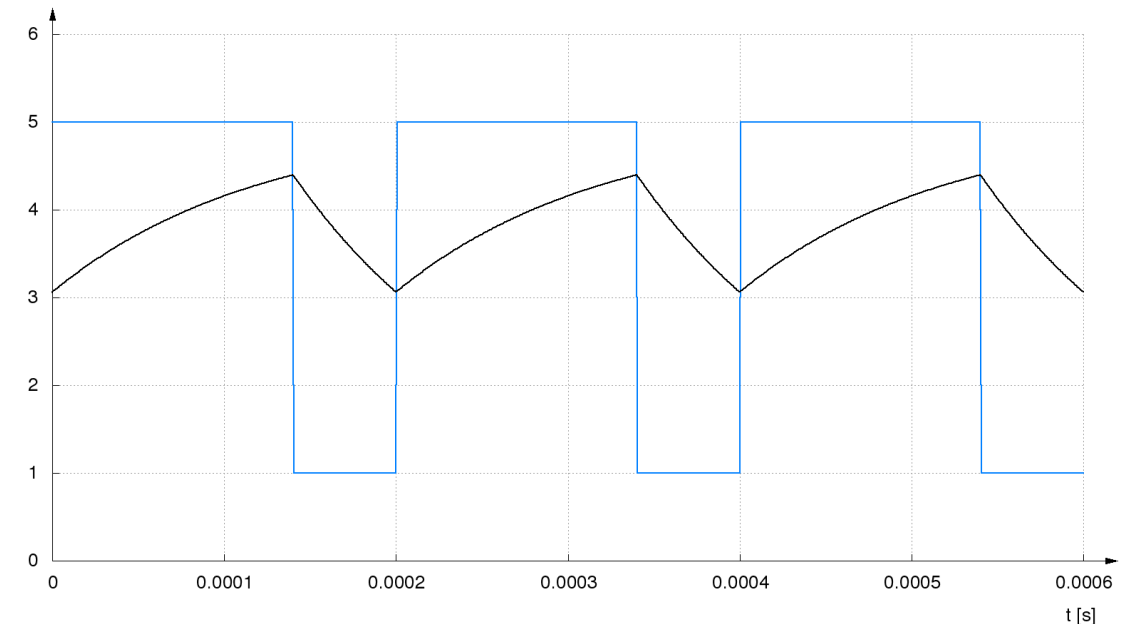
- Where $\tau = RC$

- Evaluate the waveforms at the extremes

- Determine A and B for the rising and falling cases
 - Determine the maximum and minimum voltages
 - The result is valid at the steady-state

- Example

- $R = 10k\Omega$, $C = 12nF$, $\tau = 120\mu s$
 - $V_H = 5V$, $V_L = 1V$
 - $f = 5kHz$, $T = 200\mu s$
 - $T_H = 140\mu s$, $T_L = 60\mu s$, $\delta = 70\%$



Simple low pass filter time domain response

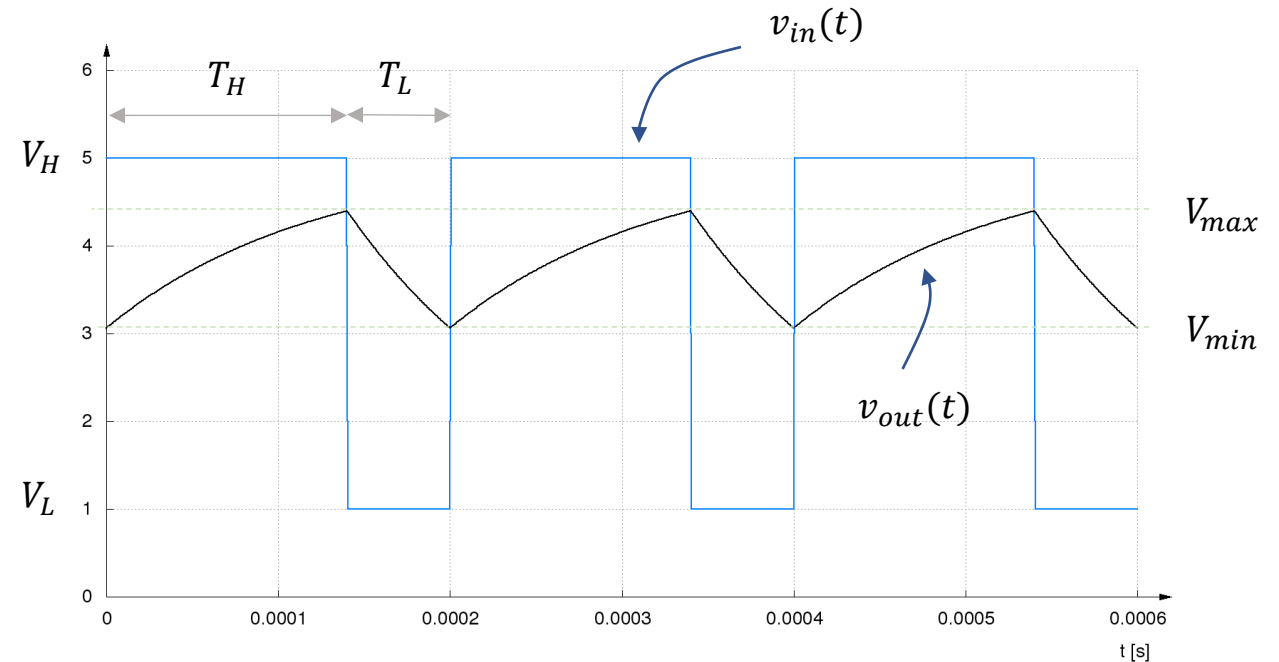
- Rising

$$\begin{cases} v_{out}^r(0) = A + B = V_{min} \\ v_{out}^r(\infty) = A = V_H \end{cases} \Rightarrow \begin{cases} A = V_H \\ B = V_{min} - V_H \end{cases} \Rightarrow v_{out}^r(t) = V_H + (V_{min} - V_H)e^{-t/\tau}$$

$$v_{out}^r(T_H) = V_H + (V_{min} - V_H)e^{-T_H/\tau} = V_{max}$$

- All terms known except

- V_{min}
- V_{max}



Simple low pass filter time domain response

- Falling

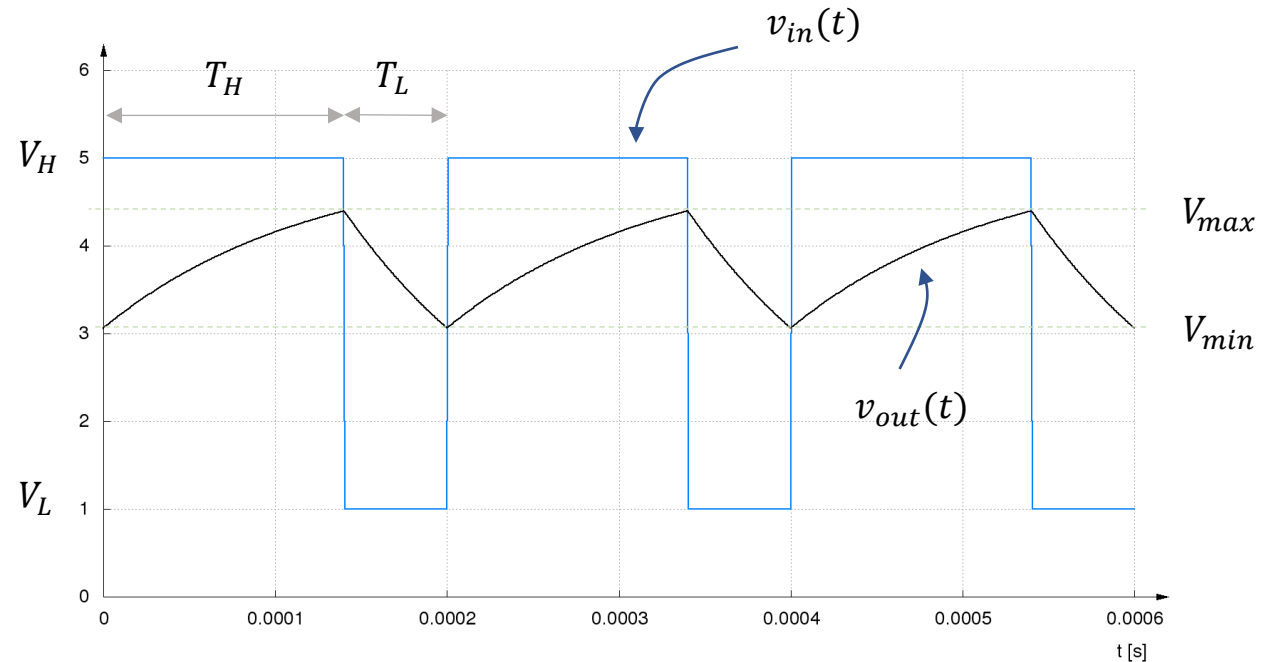
- Move time origin to T_H

$$\begin{cases} v_{out}^f(0) = A + B = V_{max} \\ v_{out}^f(\infty) = A = V_L \end{cases} \Rightarrow \begin{cases} A = V_L \\ B = V_{max} - V_L \end{cases} \Rightarrow v_{out}^f(t) = V_L + (V_{max} - V_L)e^{-t/\tau}$$

$$v_{out}^f(T_L) = V_L + (V_{max} - V_L)e^{-T_L/\tau} = V_{min}$$

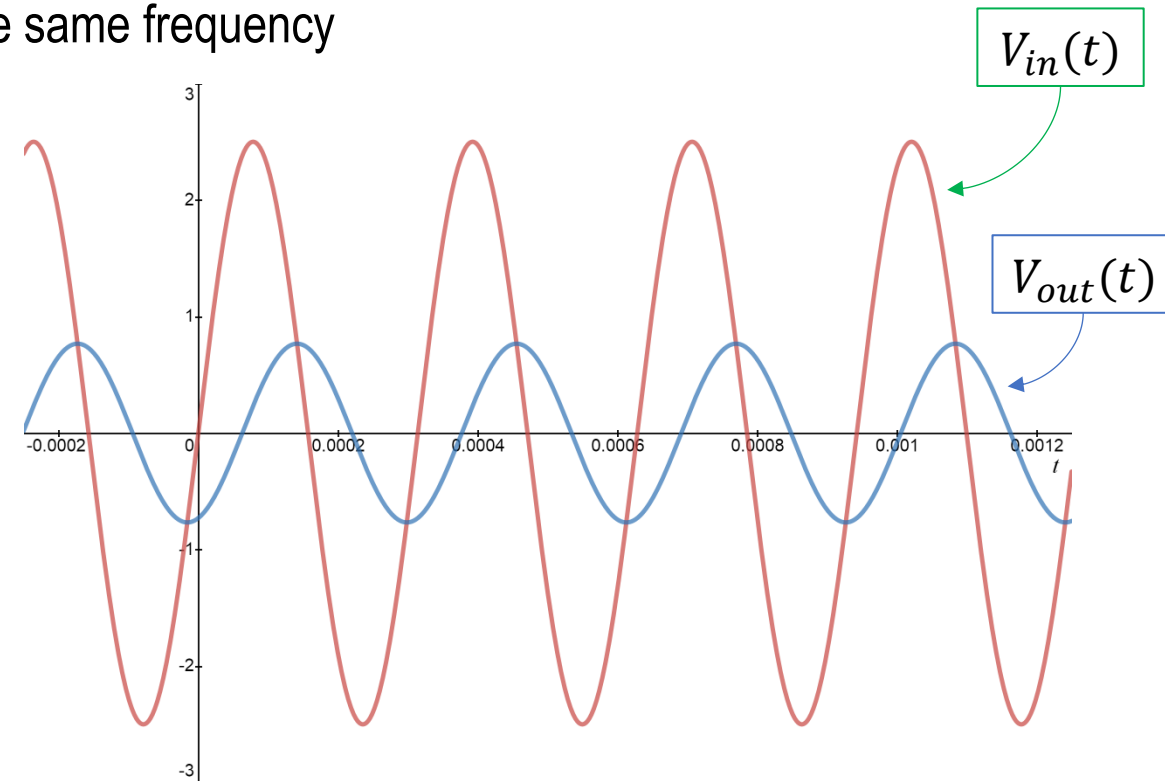
- Substituting the data

$$\begin{cases} 0.311V_{min} - V_{max} = -3.44 \\ V_{min} - 0.607V_{max} = 0.393 \end{cases}$$
$$\Rightarrow \begin{cases} V_{min} = 3.06V \\ V_{max} = 4.4V \end{cases}$$



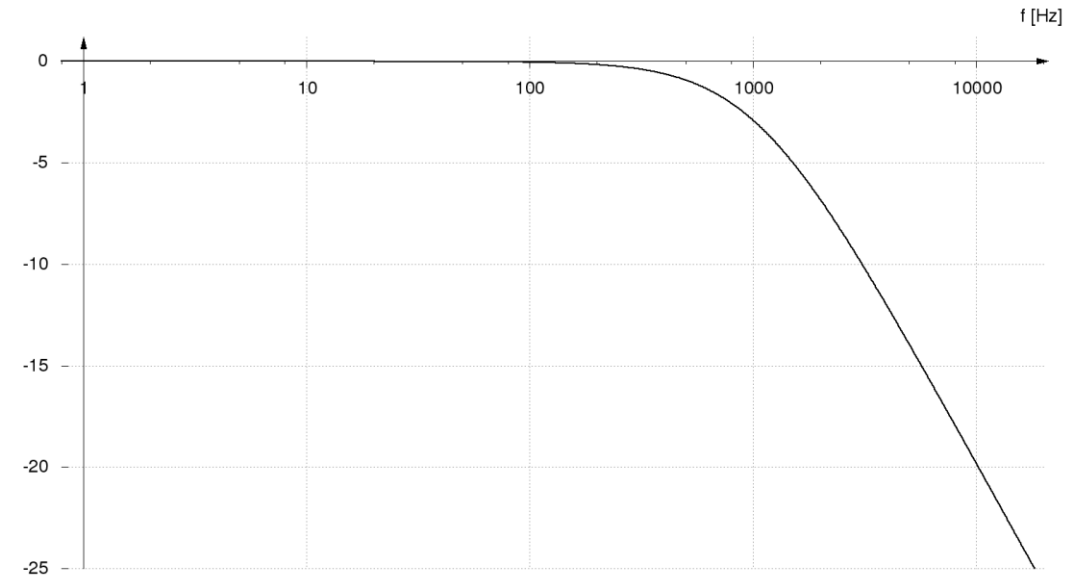
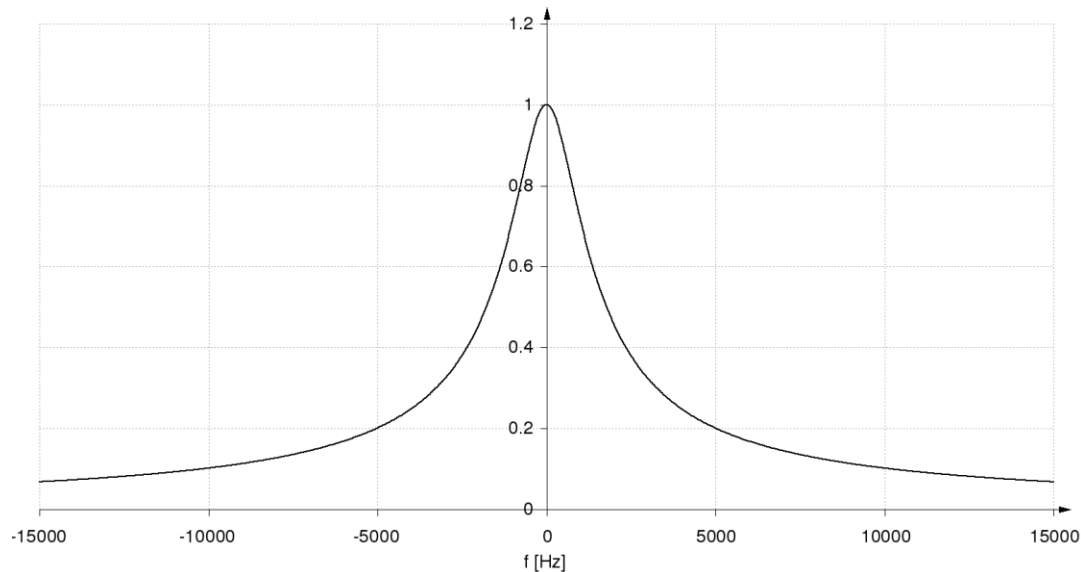
Simple low pass filter frequency response

- If V_{in} is a sinusoidal wave, compute the response
 - $V_{in}(t) = M \sin(\omega t) = M \sin(2\pi f t)$
 - In a linear system at the steady state
 - If the input is sinusoidal, the output is sinusoidal at the same frequency
 - $V_{out}(t) = M' \sin(\omega t + \phi)$
 - $M' = M \frac{\omega_p}{\sqrt{\omega^2 + \omega_p^2}}, \phi = -\tan^{-1}\left(\frac{\omega}{\omega_p}\right)$
- Example
 - $M = 2.5 \text{ V}, \omega = 20 \text{ krad/s}, f = 3183 \text{ Hz}$
 - $R = 3.3 \text{ k}\Omega, C = 47 \text{ nF}$
 - $\omega_p = \frac{1}{RC} = 6447 \text{ rad/s}, f_p = \frac{\omega_p}{2\pi} = 1026 \text{ Hz}$
 - $M' = M \cdot 0.307 = 0.767 \text{ V}$
 - $\phi = -\tan^{-1}\left(\frac{20 \cdot 10^3}{6447}\right) = -1.26 \text{ rad}$



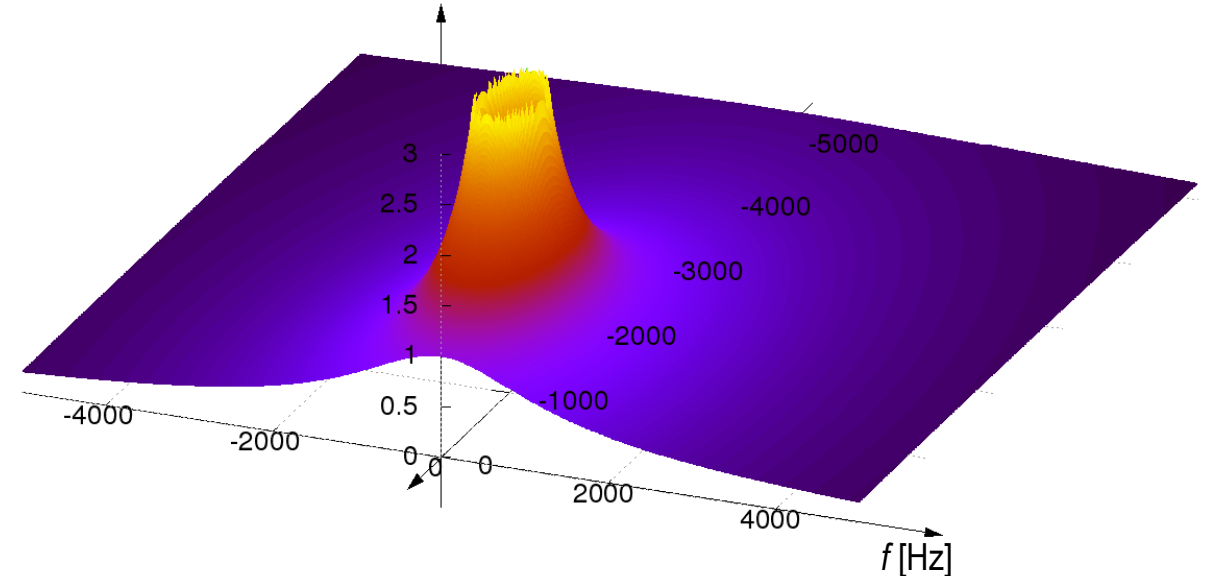
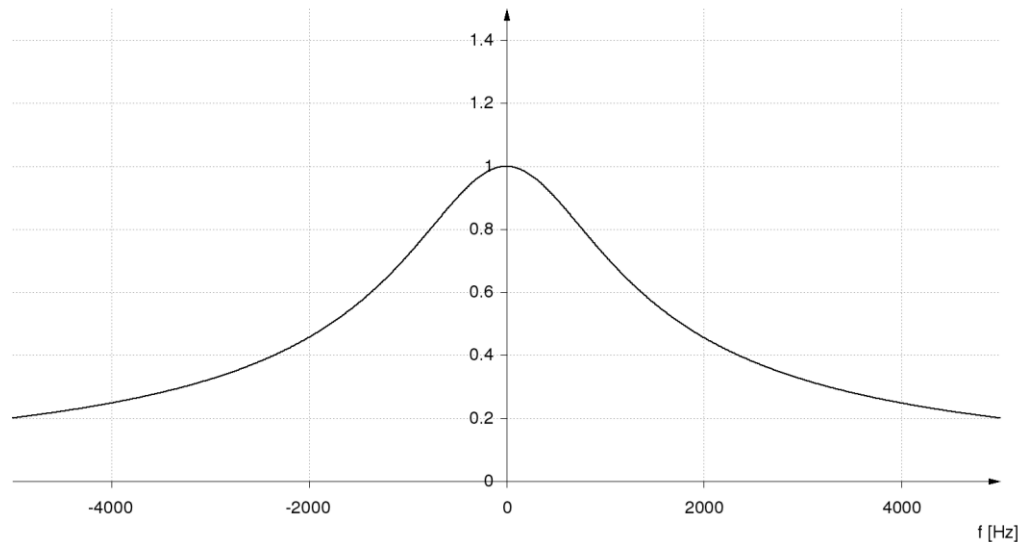
Simple low pass filter frequency response

- Transfer function graphs, linear and logarithmic (Bode) scales
 - $R = 3.3 \text{ k}\Omega$, $C = 47 \text{ nF}$, $\omega_p = \frac{1}{RC} = 6447 \text{ rad/s}$, $f_p = \frac{\omega_p}{2\pi} = 1026 \text{ Hz}$
 - Transfer function at 0.7 for $f = f_p$ in linear scale
 - Transfer function at -3dB for $f = f_p$ in log scale



Simple low pass filter frequency response

- A pole is the zero of the denominator
- Why doesn't the diagram go to infinity?
 - The pole is on the real axis
 - The Bode diagram is on the imaginary axis
 - Compare the graph in linear scales on the left with the section on the frequency axis in the picture on the right



Superposition of effects

- In a linear system with more than one input (independent ideal sources, voltage or current), the output is the algebraic sum of the effects of each input with the other set to 0
 - Voltage source at zero $\rightarrow$ short circuit
 - Current source at zero $\rightarrow$ open circuit
 - Dependent sources are NOT set to 0
- Linear system
 - Sources (independent and dependent), resistors, capacitors, inductors, transformers
- Non linear system
 - Diodes, transistors, operational amplifiers
 - Superposition of effects not applicable
 - Applied after linearization around a bias point

Millman theorem

- In the general network on the right, the voltage V_A is given by the expression

$$V_A = \frac{\sum_{i=1}^n \frac{V_i}{R_i} - \sum_{j=1}^m I_j}{\sum_{i=1}^n \frac{1}{R_i}}$$

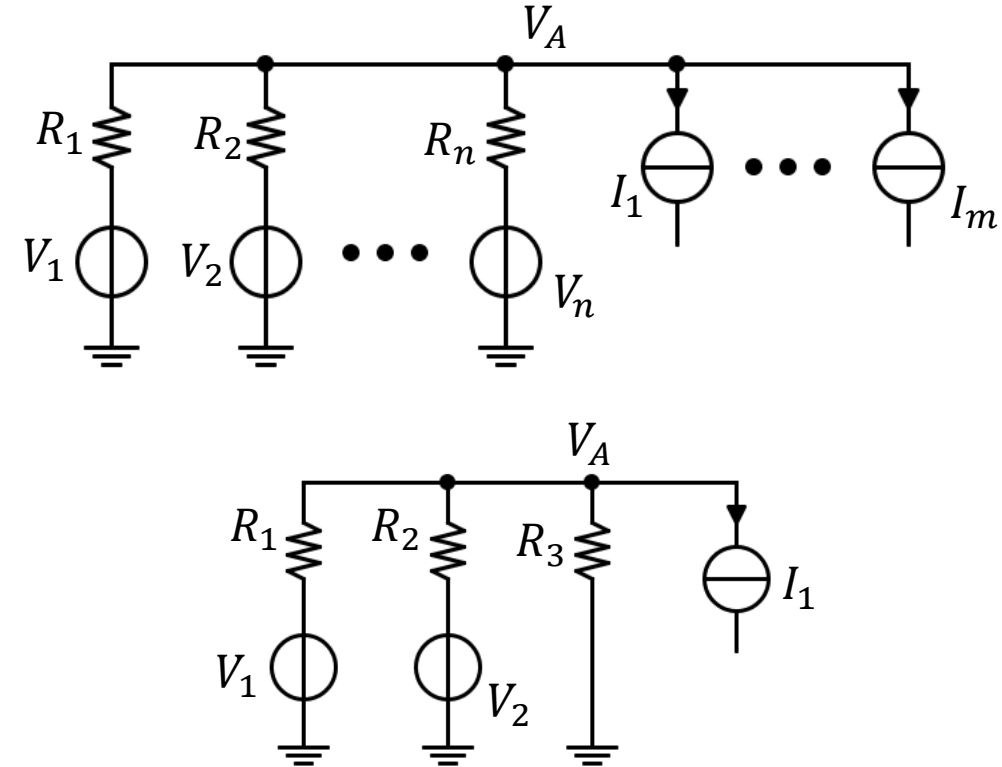
- Example

$$V_A = \frac{\frac{V_1}{R_1} + \frac{V_2}{R_2} - I_1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

- Note that $V_3 = 0$

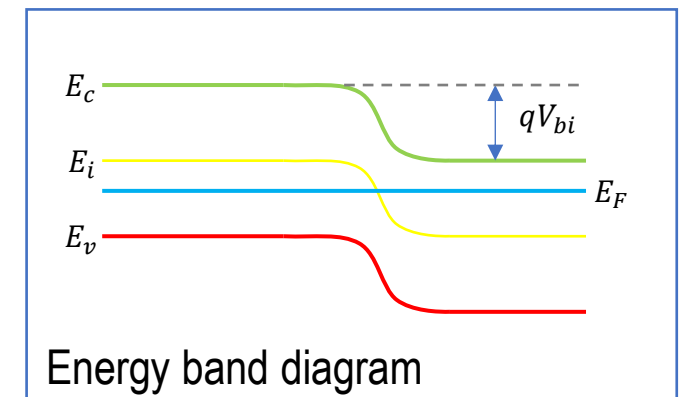
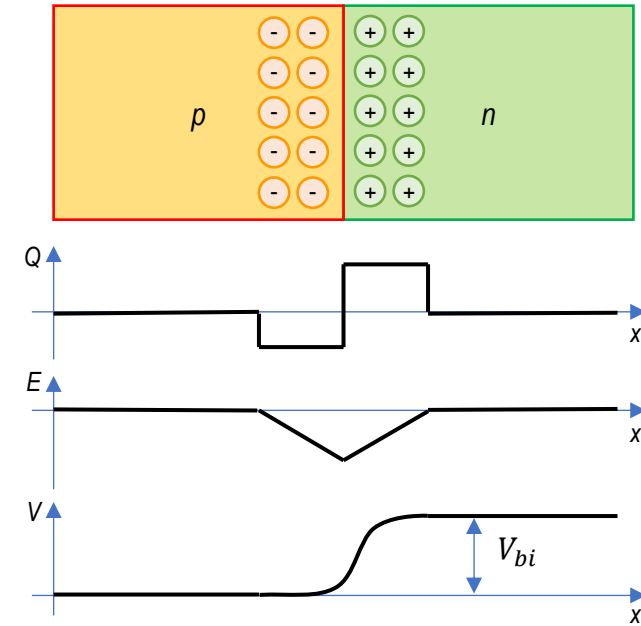
- Homework

- Prove the Millman theorem using the superposition of effects and voltage dividers with conductances



Diode

- Non linear component
- p-n junction
 - Initial phase
 - Electrons from the n side flow by diffusion to the p side
 - Fixed positive ions are left in the n side
 - Holes from the p side flow by diffusion to the n side
 - Fixed negative ions are left in the p side
 - At thermal equilibrium
 - Fixed ions in the two sides create an electrical field which opposes the further diffusion of electrons and holes
 - *Depletion Layer* or *Space Charge Region*
 - A built-in voltage V_{bi} is established
 - No current flows in the diode
 - The built-in voltage is a barrier



Diode

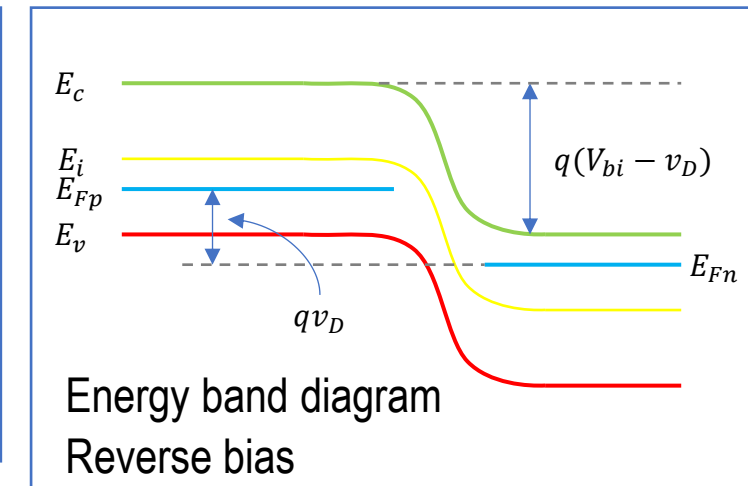
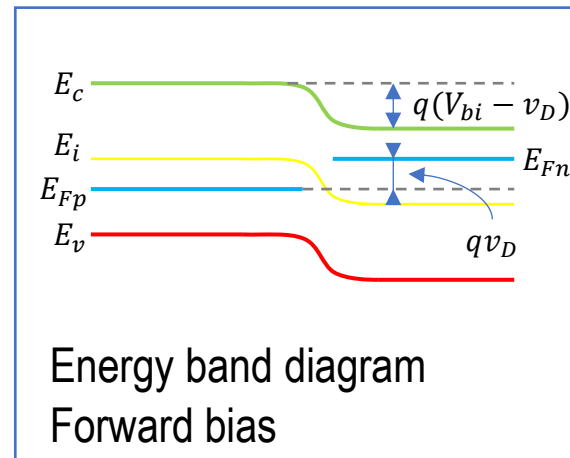
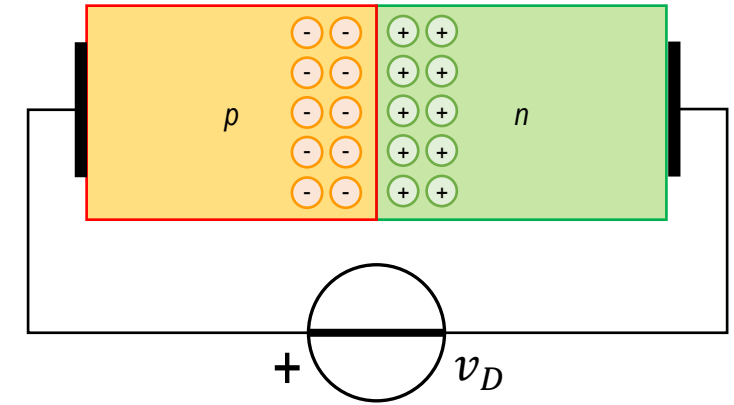
- Applying a voltage to the junction

- Forward bias

- No current flows if $v_D < V_{bi}$
 - Current flows when $v_D > V_{bi}$
 - $V_{bi} \approx 0,5V$ in normal small signal diodes

- Reverse bias

- No current flows
 - Depletion layer width increases
 - For large reverse voltages
 - Breakdown of the junction
 - Large reverse current
 - Tunneling effect or avalanche effect
 - No damages to the diode
 - The large current may increase power dissipation and heat, causing damages
 - Exploited in Zener diodes



Diode

- I-V Characteristic:

- $i_D(t) = I_S \left(e^{\frac{v_D(t)}{\eta V_T}} - 1 \right)$

- I_S : reverse current

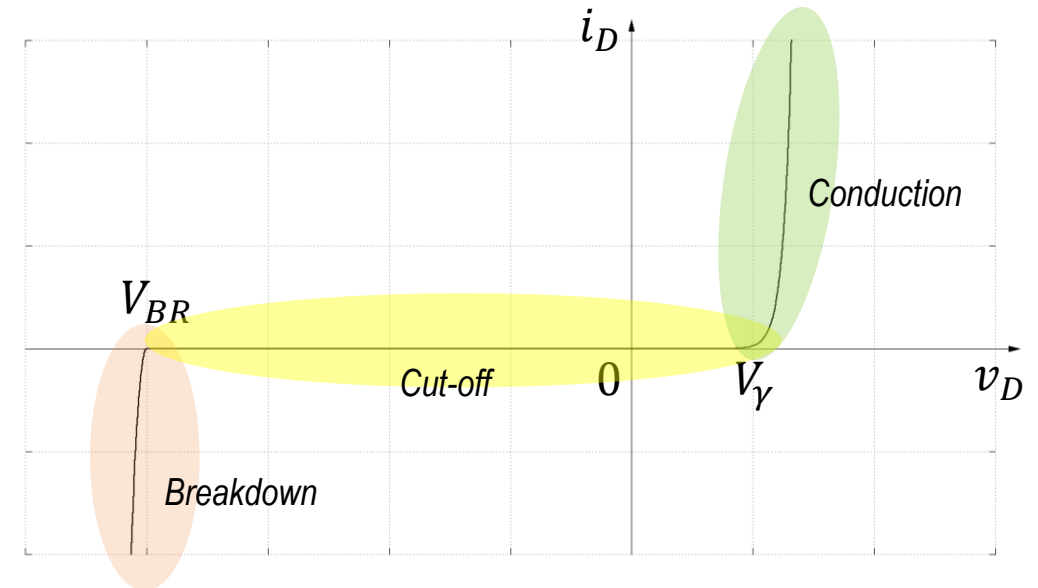
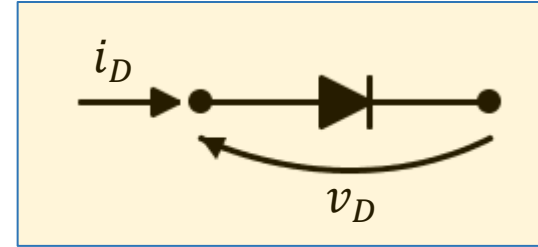
- Very small, in the order of nA or less

- η : a value between 1 and 2

- Assume $\eta = 1$ in the following

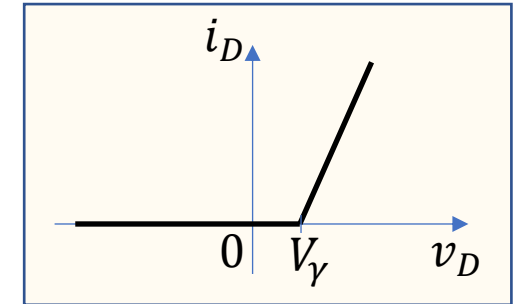
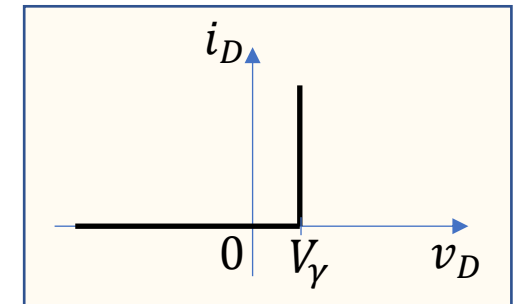
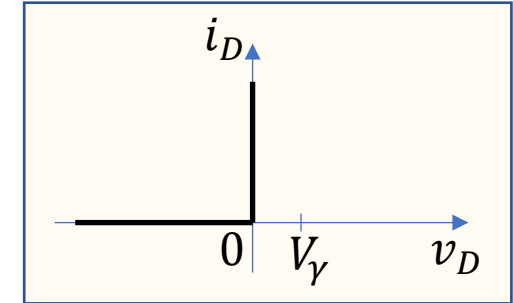
- $V_T = \frac{KT}{q}$: voltage equivalent of temperature

- T : Temperature in Kelvin
 - $K = 1.38 \cdot 10^{-23} \text{ m}^2 \text{ kg s}^{-1} \text{ K}^{-1}$: Boltzmann constant
 - $q = 1.602 \cdot 10^{-19}$ coulombs: charge of an electron
 - $V_T = 26 \text{ mV}$ at room temperature (300K)



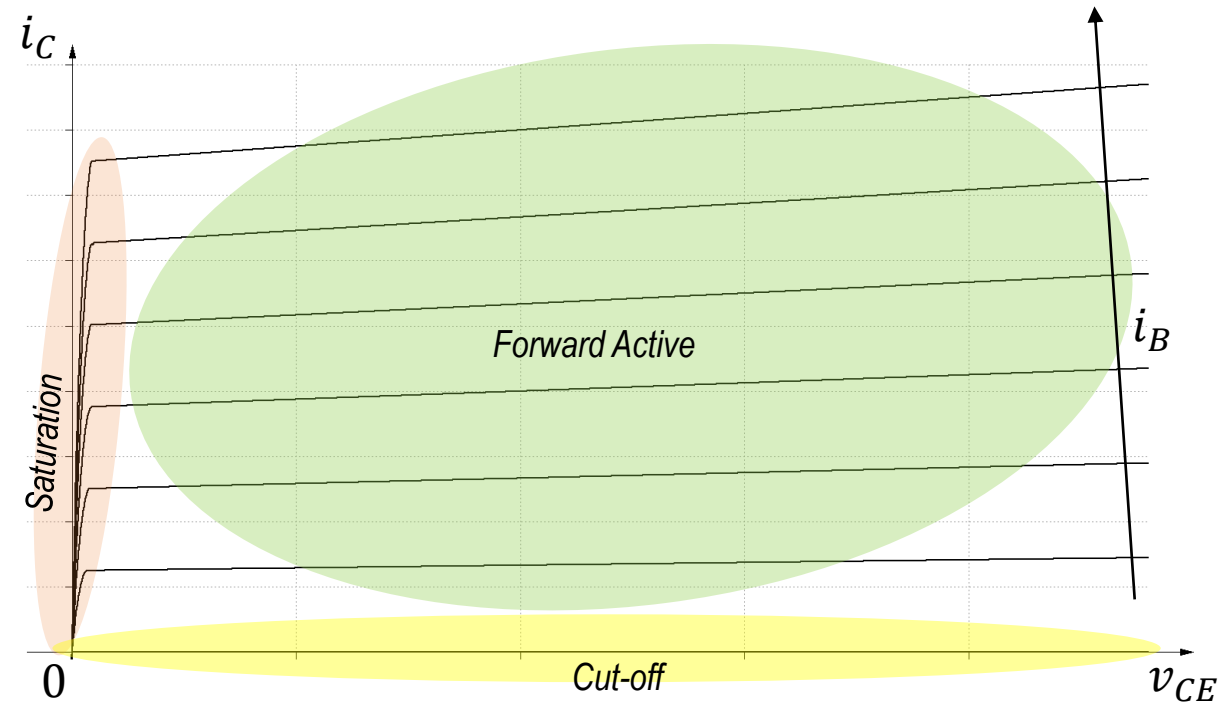
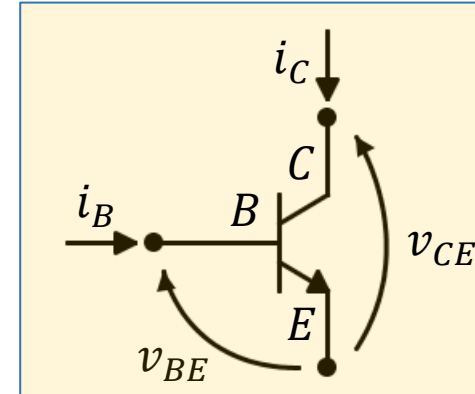
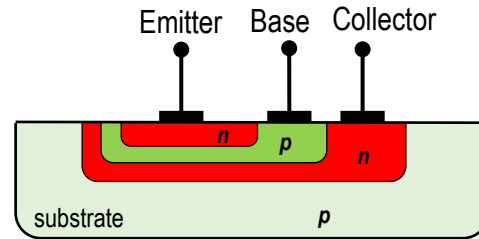
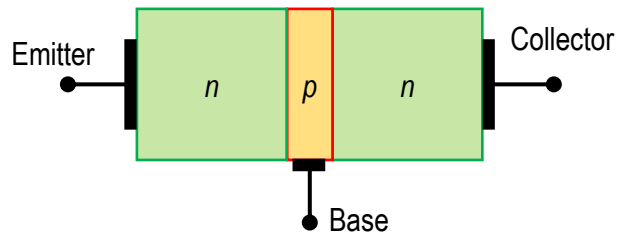
Diode

- Approximate models
 - Ideal switch
 - When voltage drop is negative, the current is 0
 - When current is positive, the voltage drop is 0
 - Ideal switch with offset
 - When voltage drop is less than V_γ , the current is 0
 - When current is positive, the voltage drop is V_γ
 - Ideal switch with dynamic resistance
 - When voltage drop is less than V_γ , the current is 0
 - When voltage drop is more than V_γ , the current follows a resistive law
- Use the ideal switch model in most cases
 - Use the ideal switch with offset model in some cases



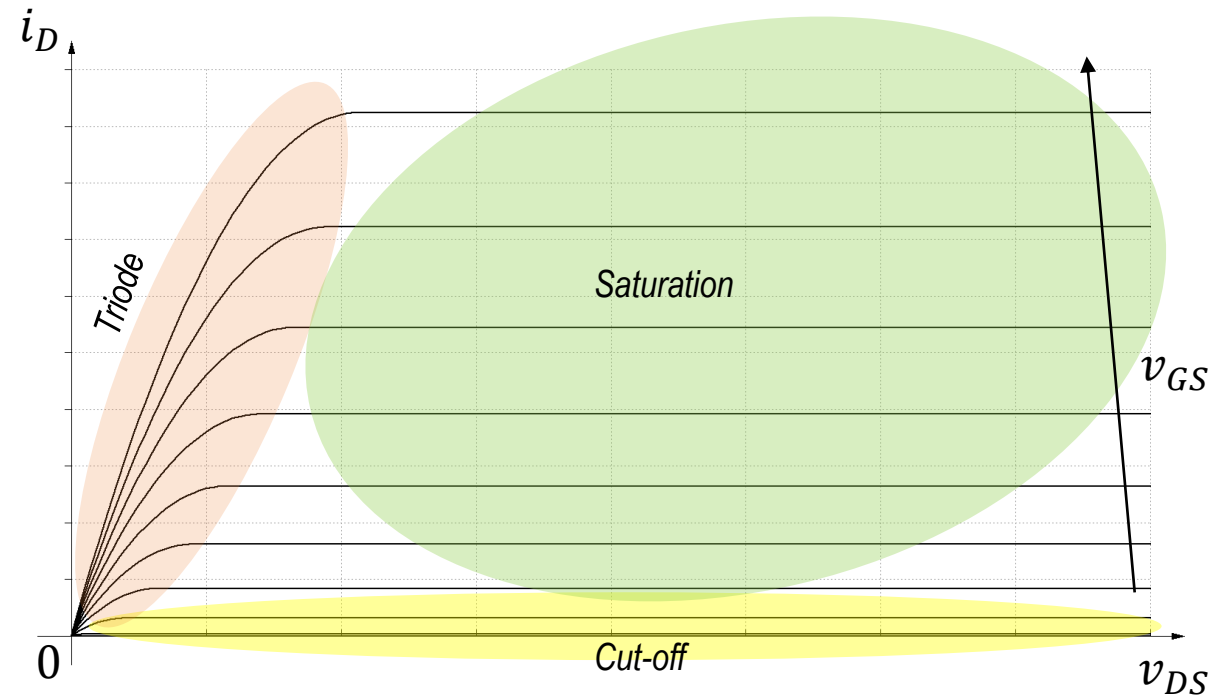
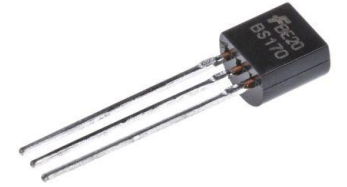
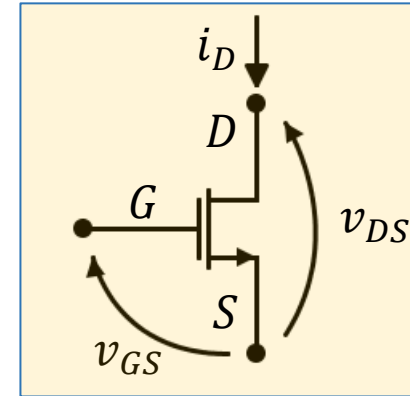
BJT transistor (npn)

- Non linear component
 - In forward active region
 - $i_C = \beta i_B$
 - $v_{BE} \approx 0.7\text{ V}$
 - In cut-off region
 - $i_C = 0$
 - Open switch
 - In saturation region
 - $v_{CE} \approx 0$
 - Closed switch



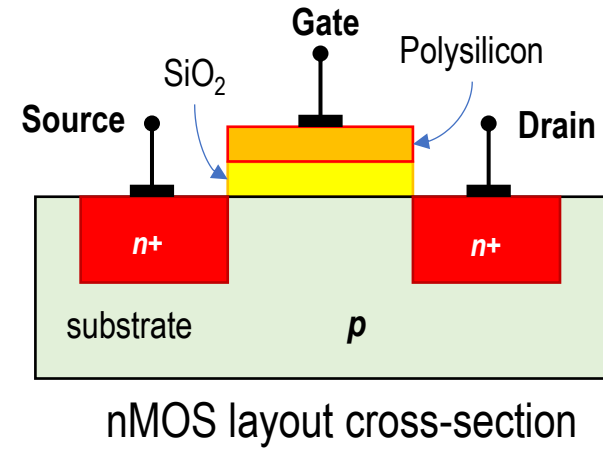
MOS transistor (nMOS)

- Non linear component
 - Threshold voltage v_{th}
 - In cut-off region
 - $v_{GS} < v_{th}, i_D = 0$
 - Open switch
 - In triode region
 - $v_{GS} > v_{th}, v_{DS} < v_{GS} - v_{th}$
 - $i_D = \mu C_{ox} \frac{W}{L} \left[(v_{GS} - v_{th}) v_{DS} - \frac{v_{DS}^2}{2} \right]$
 - Closed switch, $v_{DS} \simeq 0$
 - In saturation region
 - $v_{GS} > v_{th}, v_{DS} > v_{GS} - v_{th}$
 - $i_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (v_{GS} - v_{th})^2$



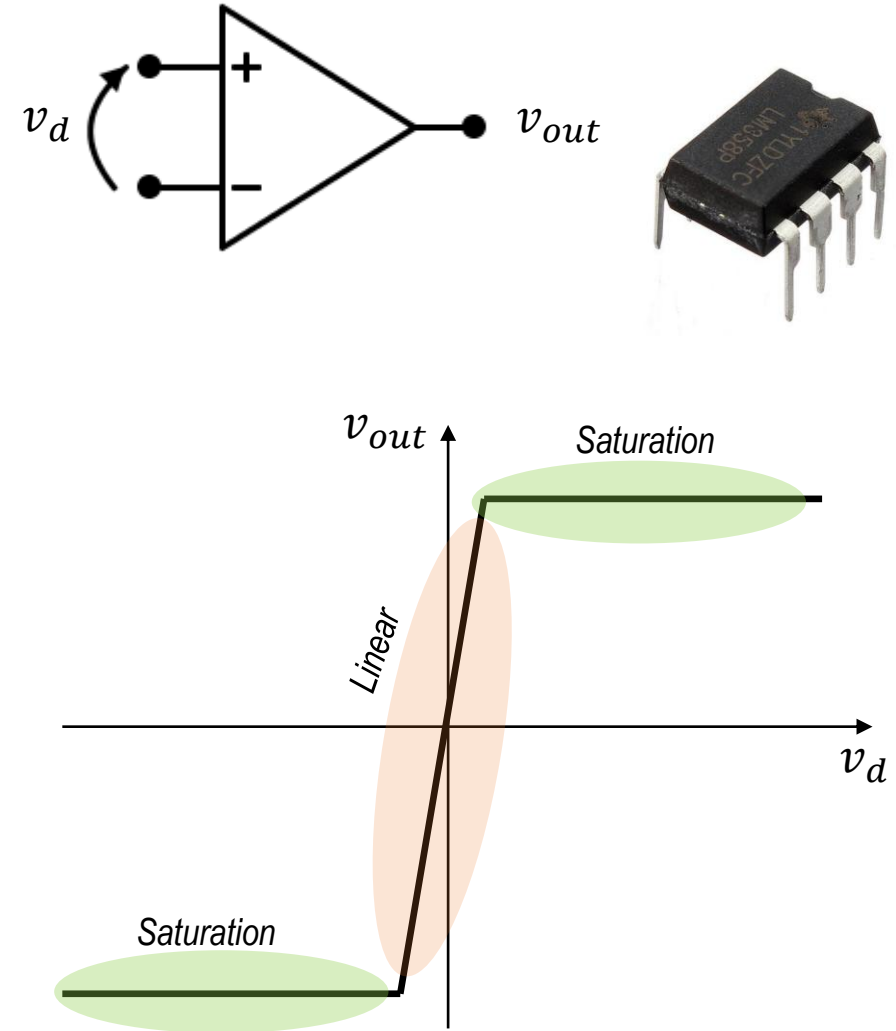
MOS transistor (nMOS)

- Non linear component
 - In deep triode region v_{DS} is small
 - $i_D = \mu C_{ox} \frac{W}{L} (v_{GS} - v_{th}) v_{DS}$
 - Linear relationship between voltage and current
 - It behaves as a resistance of value
 - $R_{DSon} = \frac{v_{DS}}{i_D} = \frac{1}{\mu C_{ox} \frac{W}{L} (v_{GS} - v_{th})}$
 - It is often a small resistance
 - Listed in the device datasheet



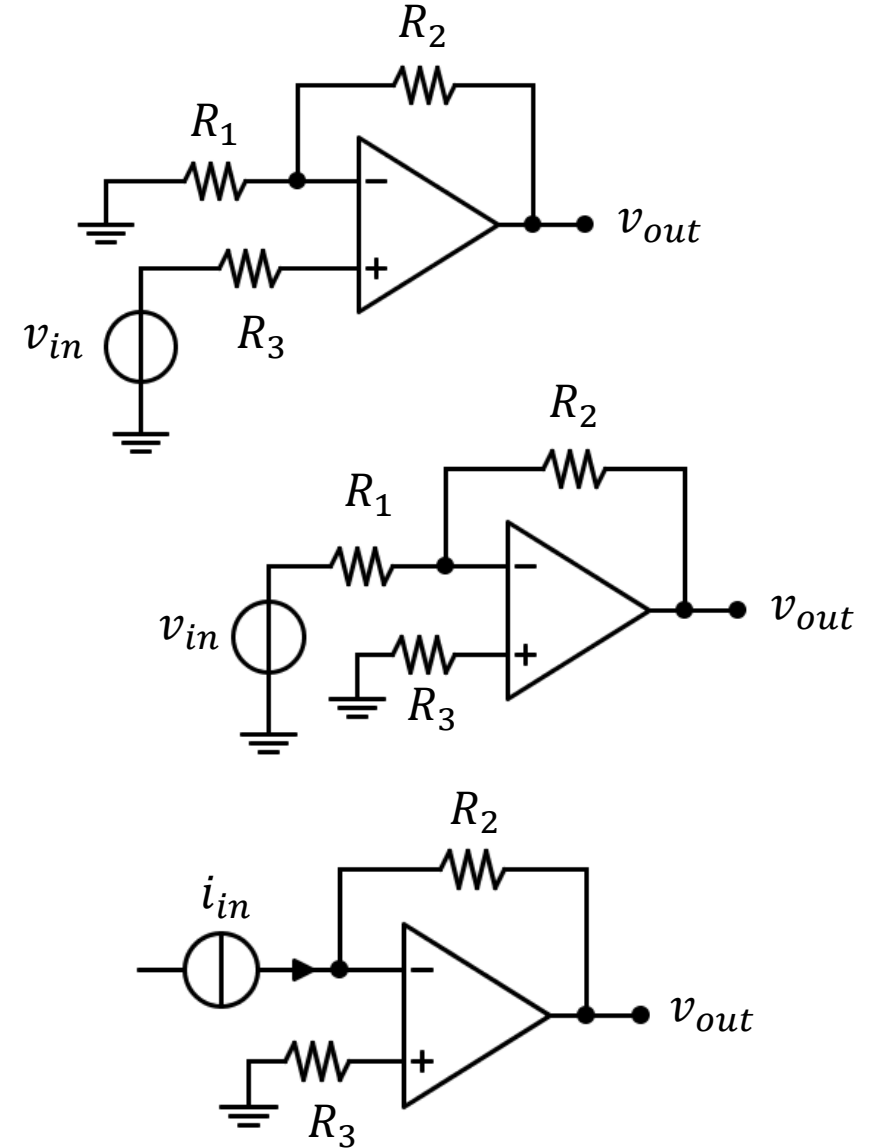
Operational amplifiers

- Differential amplifier with very high gain
 - $v_{out} = A_d(v^+ - v^-) = A_d v_d$
 - A_d in the order of several hundreds thousands or millions
- Used as a linear amplifier
 - With a negative feedback network
 - Forced to work in linear region
 - Lower final gain, larger bandwidth, stability
- Used as a comparator
 - With a positive feedback network (or no feedback)
 - Forced to work in saturation regions
- Widely used in many embedded systems



Operational amplifiers

- Non inverting amplifier
 - $v_{out} = v_{in} \left(1 + \frac{R_2}{R_1}\right)$, $R_{in} = \infty$, $R_{out} = 0$
 - R_3 used to reduce unwanted output offsets
- Inverting amplifier
 - $v_{out} = v_{in} \left(-\frac{R_2}{R_1}\right)$, $R_{in} = R_1$, $R_{out} = 0$
 - Note the non ideal (low) input resistance
- Transresistance amplifier
 - $v_{out} = -R_2 i_{in}$, $R_{in} = 0$, $R_{out} = 0$



Operational amplifiers

- Generalized analog adder/subtractor

- We want to get

$$v_{out} = A'_1 v'_1 + \dots + A'_n v'_n - (A''_1 v''_1 + \dots + A''_m v''_m)$$

- Fix the value of the feedback resistance R_f
- All coefficients with negative sign can be obtained by setting

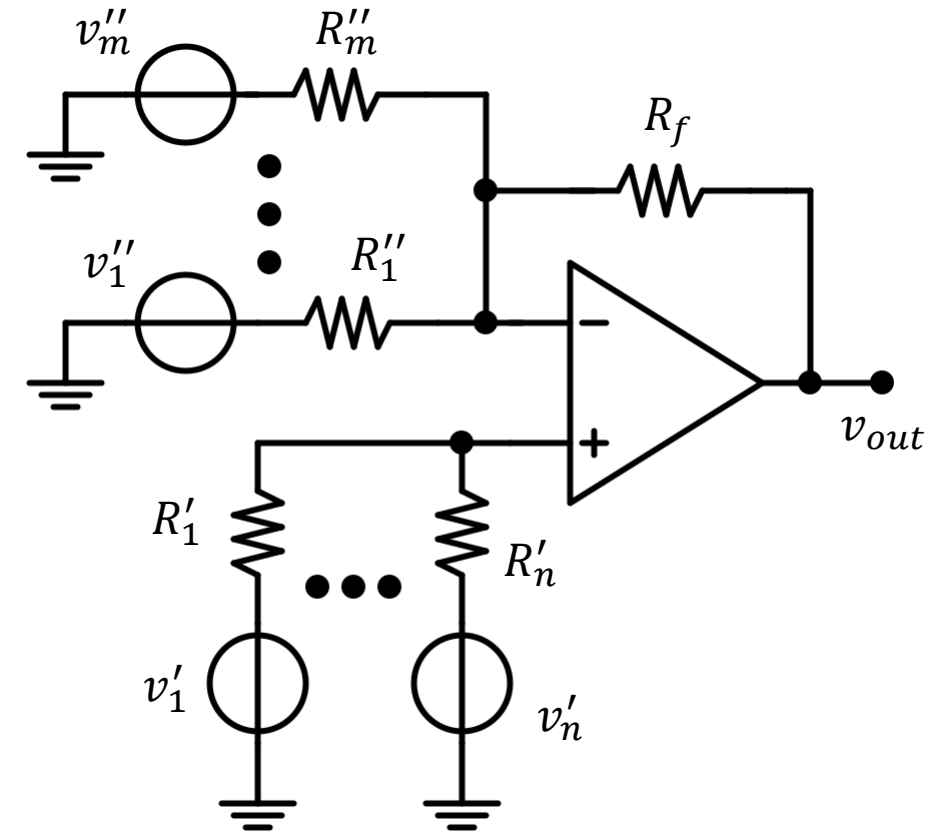
$$R''_i = \frac{R_f}{A''_i} \quad \text{or} \quad g''_i = \frac{A''_i}{R_f}$$

- The voltage on the $+$ terminal is obtained using the Millmann theorem

$$V_+ = \frac{\frac{v'_1}{R'_1} + \frac{v'_2}{R'_2} + \dots + \frac{v'_n}{R'_n}}{\frac{1}{R'_1} + \frac{1}{R'_2} + \dots + \frac{1}{R'_n}} = \frac{g'_1 v'_1 + g'_2 v'_2 + \dots + g'_n v'_n}{g'_1 + g'_2 + \dots + g'_n} = \frac{\sum_n g'_j v'_j}{\sum_n g'_j}$$

- Setting

$$R'_j = \frac{R_f}{A'_j} \quad \text{or} \quad g'_j = \frac{A'_j}{R_f} \quad \Rightarrow \quad V_+ = \frac{\sum_n A'_j v'_j}{\sum_n A'_j}$$



Operational amplifiers

- Generalized analog adder/subtractor

- We want to get

$$v_{out} = A'_1 v'_1 + \dots + A'_n v'_n - (A''_1 v''_1 + \dots + A''_m v''_m)$$

- Set

$$R''_i = \frac{R_f}{A''_i}$$

$$R'_j = \frac{R_f}{A'_j}$$

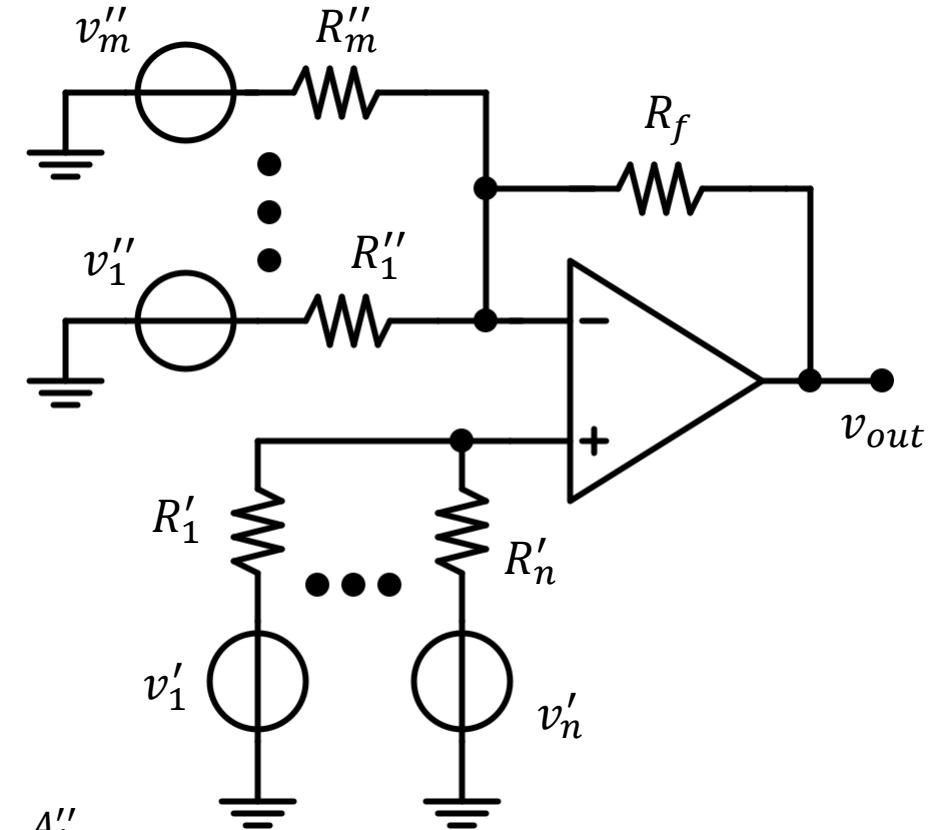
$$V_+ = \frac{\sum_n A'_j v'_j}{\sum_n A'_j}$$

- The V_+ voltage is amplified by a quantity

$$k = 1 + \frac{R_f}{R''_1 \parallel R''_2 \parallel \dots \parallel R''_m} = 1 + \sum_m g''_i R_f = 1 + \sum_m A''_i$$

- For the amplifier to work we want

$$kV_+ = \sum_n A'_j v'_j \Rightarrow \left(1 + \sum_m A''_i\right) \frac{\sum_n A'_j v'_j}{\sum_n A'_j} = \sum_n A'_j v'_j \Rightarrow \frac{1 + \sum_m A''_i}{\sum_n A'_j} = 1$$



Operational amplifiers

- Generalized analog adder/subtractor

- We want to get

$$v_{out} = A'_1 v'_1 + \dots + A'_n v'_n - (A''_1 v''_1 + \dots + A''_m v''_m)$$

- Set

$$R''_i = \frac{R_f}{A''_i}$$

$$R'_j = \frac{R_f}{A'_j}$$

$$V_+ = \frac{\sum_n A'_j v'_j}{\sum_n A'_j}$$

- We want

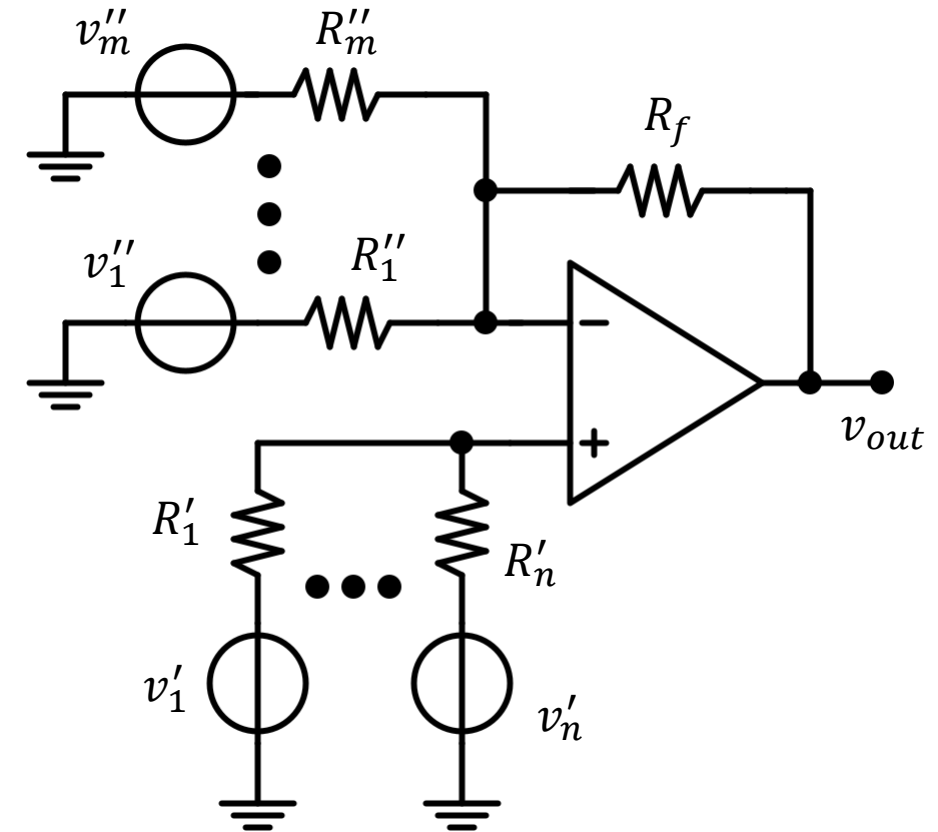
$$\frac{1 + \sum_m A''_i}{\sum_n A'_j} = 1$$

- If it is greater than 1, then add a resistance on the + terminal

$$A'_{n+1} = 1 + \sum_m A''_i - \sum_n A'_j \quad R'_{n+1} = \frac{R_f}{A'_{n+1}} = \frac{R_f}{1 + \sum_m A''_i - \sum_n A'_j}$$

- If it is less than 1, then add a resistance on the - terminal

$$A''_{m+1} = \sum_n A'_j - 1 - \sum_m A''_i \quad R''_{m+1} = \frac{R_f}{A''_{m+1}} = \frac{R_f}{\sum_n A'_j - 1 - \sum_m A''_i}$$



Operational amplifiers

- Homework

- $v_{out} = 2v'_1 + 3v'_2 - (2v''_1 + 4v''_2)$

- $\frac{1 + \sum_m A''_i}{\sum_n A'_j} = \frac{7}{5}$

- Greater than 1, add a resistance on the + terminal such that $A'_3 = 2$ with $v'_3 = 0$ (a short circuit)

- $R'_3 = R_f/2$

- $v_{out} = 2v'_1 + 6v'_2 - (2v''_1 + 3v''_2)$

- $v_{out} = 2v'_1 + 4v'_2 - (2v''_1 + 3v''_2)$

- $v_{out} = 12v'_1 + 5V$

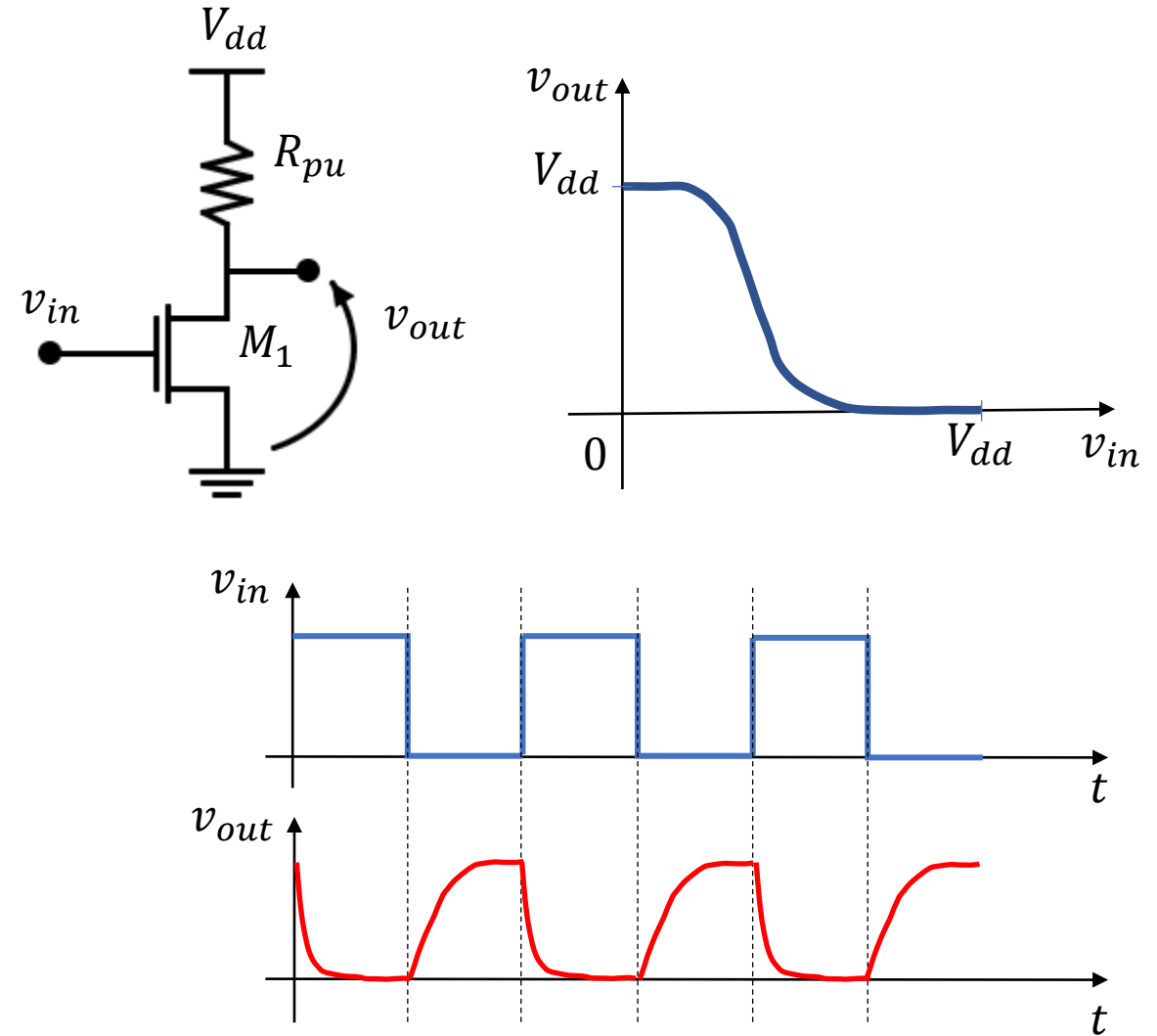
- How to choose R_f

- Not too small, otherwise too much current in the feedback network
 - Not too high, otherwise bias currents do not flow
 - Usually tens or hundreds of k Ω

Logic gates

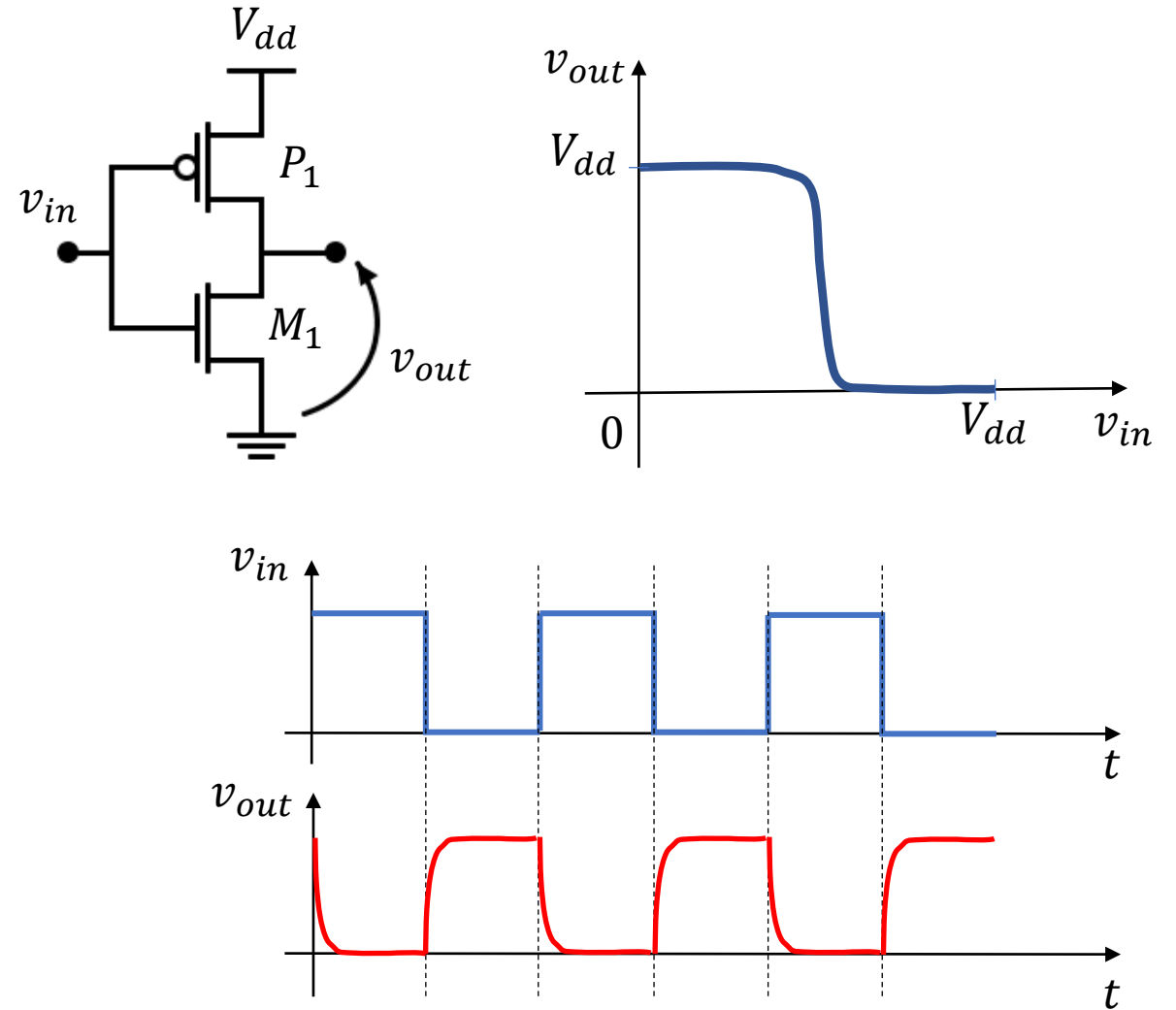
- Inverter

- nMOS inverter with pull-up resistor
 - Asymmetric response
 - Falling edge much sharper than rising edge
 - R_{DSon} of nMOS much smaller than R_{pu}
 - Static power consumption
 - When output is at logic 0
 - $P = \frac{V_{dd}^2}{R_{pu} + R_{DSon}}$
 - Dynamic power consumption



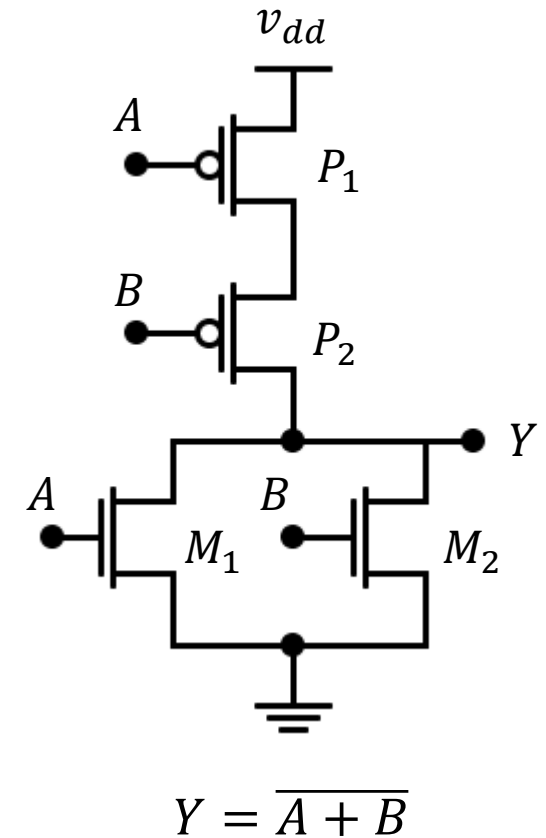
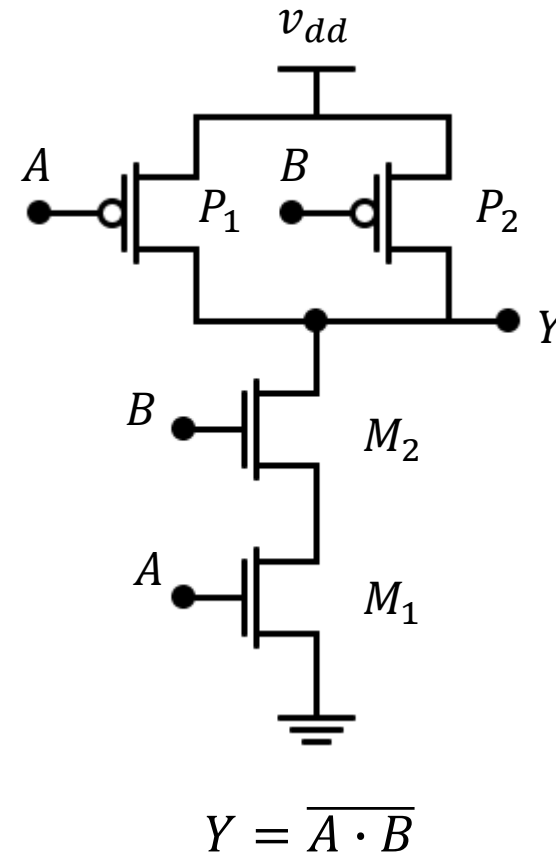
Logic gates

- Inverter
 - CMOS inverter
 - Symmetric response
 - If pMOS is larger than nMOS
 - No static power consumption
 - Dynamic power consumption
 - $P \propto CfV^2$
 - CMOS technology
 - Very common in many integrated circuits



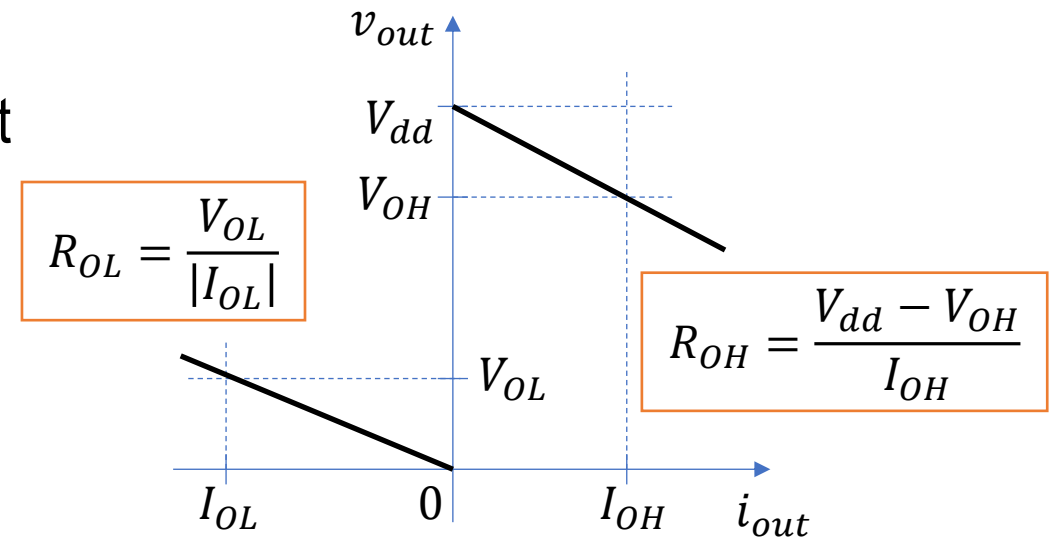
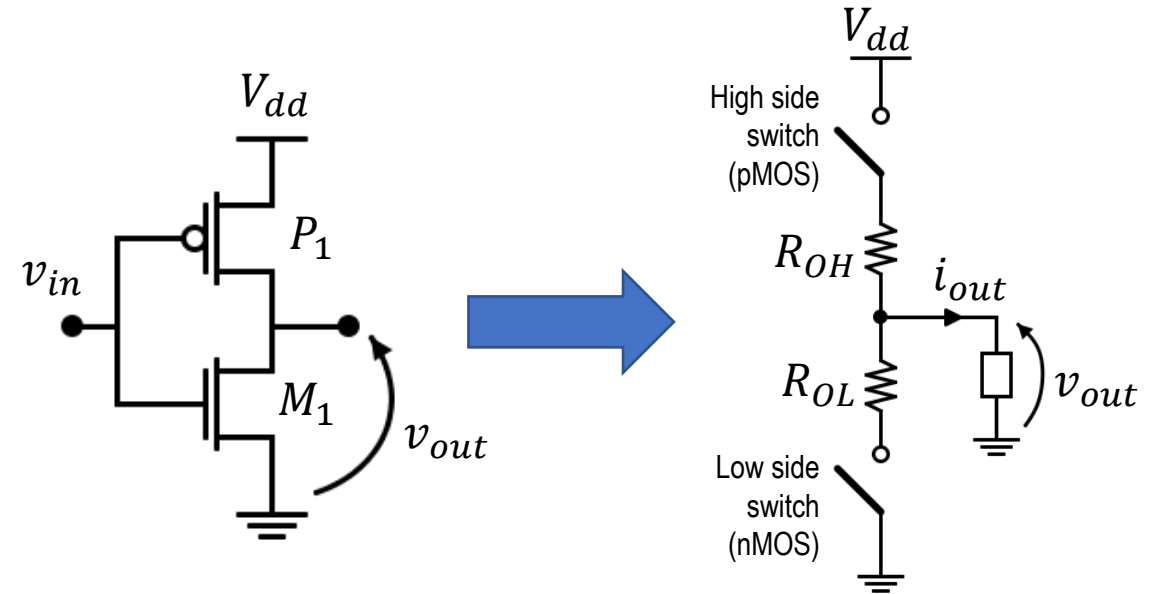
Logic gates

- Other logic functions
 - Use a combination of transistor in series or in parallel
 - Pull-down branch
 - Series implements a NAND
 - Parallel implements a NOR
 - Pull-up branch
 - Series implements a NOR
 - Parallel implements a NAND
 - For affirmative functions, put an inverter at the output



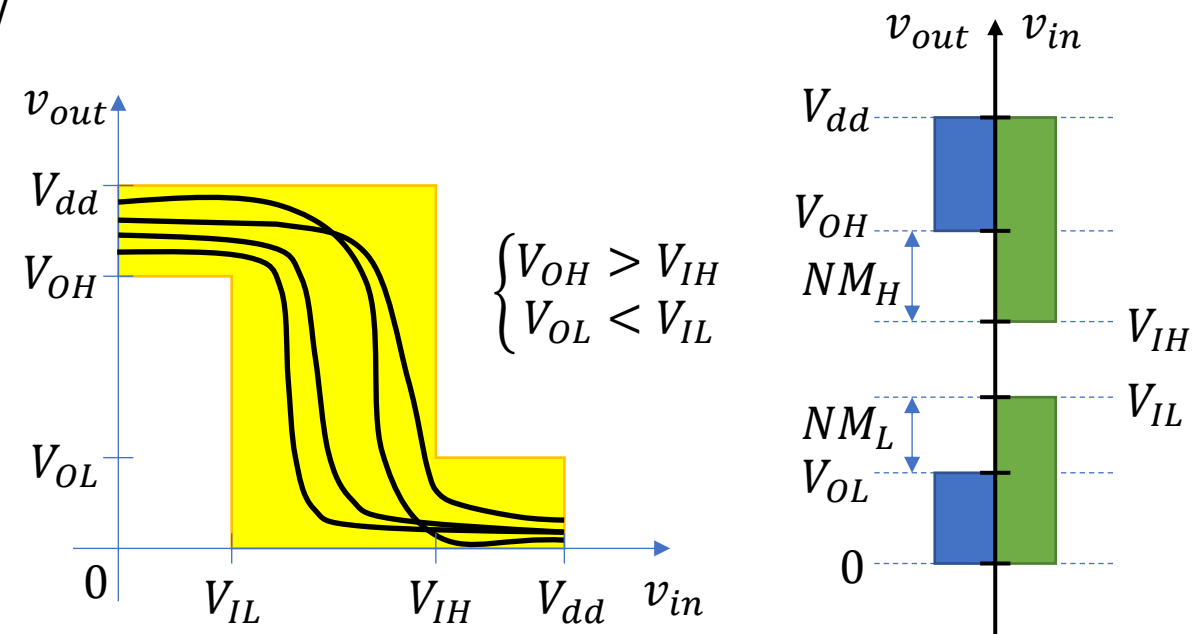
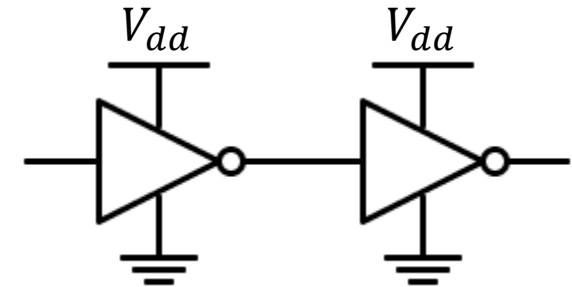
Logic gates

- Logic gate output parameters
 - Model each transistor as a switch and a series resistance
 - V_{OH} : minimum output voltage at logic high
 - V_{OL} : maximum output voltage at logic low
 - I_{OH} : maximum output current such that the output voltage at logic high is greater than V_{OH}
 - I_{OL} : maximum output current such that the output voltage at logic low is less than V_{OL}



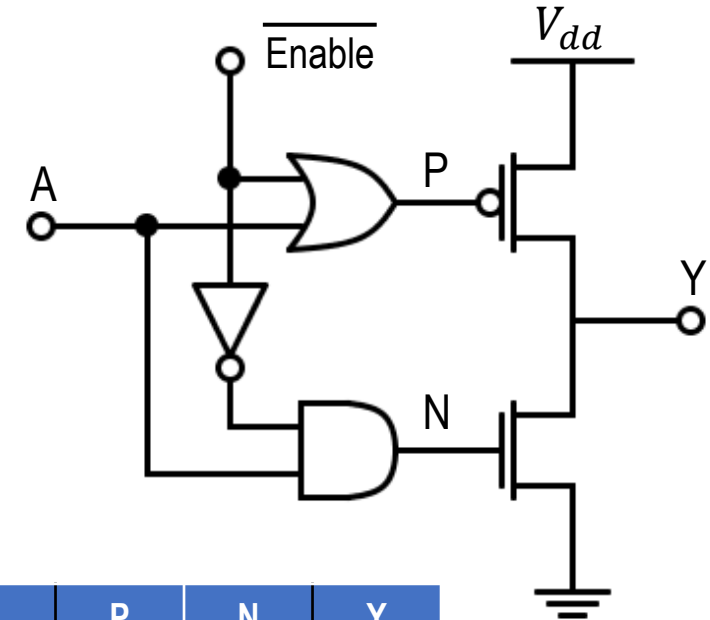
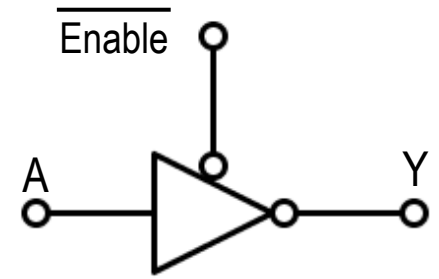
Logic gates

- Logic gate compatibility
 - Logic gate input parameters
 - V_{IH} : minimum input voltage recognized as logic high
 - V_{IL} : maximum input voltage recognized as logic low
 - Statistical parameters
 - Consider thousands of units of the same logic family
 - Find the maximum of all the V_{IH}
 - This is the V_{IH} for the entire family
 - Find the minimum of all the V_{IL}
 - This is the V_{IL} for the entire family
 - Conditions
 - $V_{OH} > V_{IH}$ by some noise margin NM_H
 - $V_{OL} < V_{IL}$ by some noise margin NM_L



Logic gates

- Tri-state (high impedance) output
 - Additional Enable input
 - Active high or active low
 - When Enable is active
 - The output is driven normally
 - When Enable is NOT active
 - Both the pull-up (high side) and pull-down (low side) transistors are switched OFF
 - The output line is floating
 - The voltage level (logic level) is set by some other circuit...
 - ... or it remains not determined
 - It can assume any voltage value!
 - May cause damages

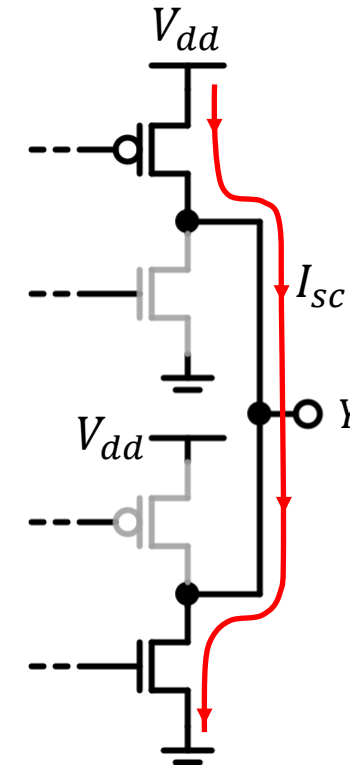
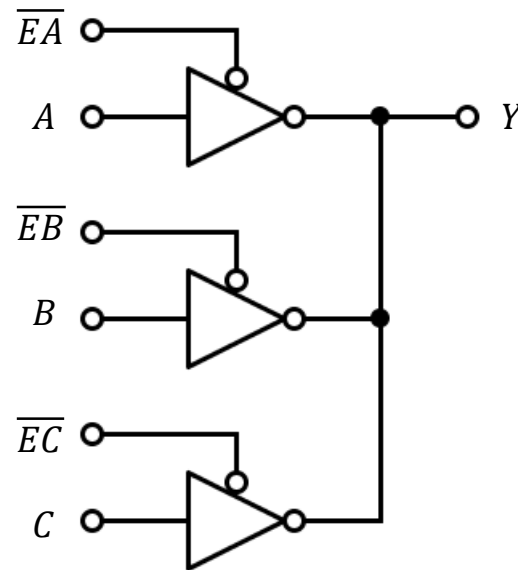


$\overline{\text{Enable}}$	A	P	N	Y
0	0	0	0	1
0	1	1	1	0
1	0	1	0	Z
1	1	1	0	Z

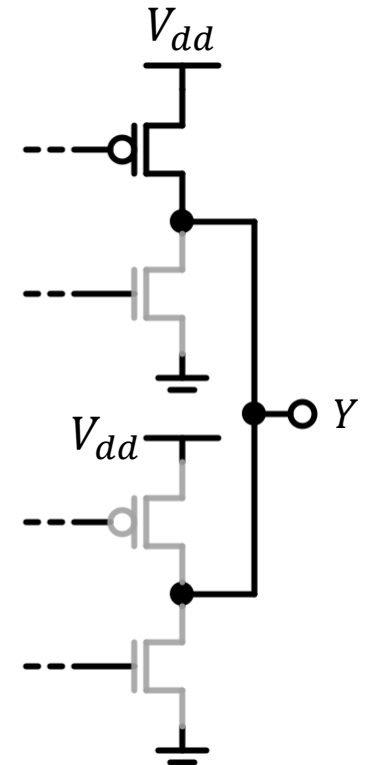
Logic gates

- Tri-state (high impedance) output
 - Useful when multiple logic gates drive the same line
 - Example: shared bus
 - All drivers are in high impedance except at most one
 - Need to guarantee the above condition to avoid short circuits

$\overline{EC}$	$\overline{EB}$	$\overline{EA}$	Y
0	0	0	Possible conflict
0	0	1	Possible conflict
0	1	0	Possible conflict
0	1	1	$\overline{C}$
1	0	0	Possible conflict
1	0	1	$\overline{B}$
1	1	0	$\overline{A}$
1	1	1	Z



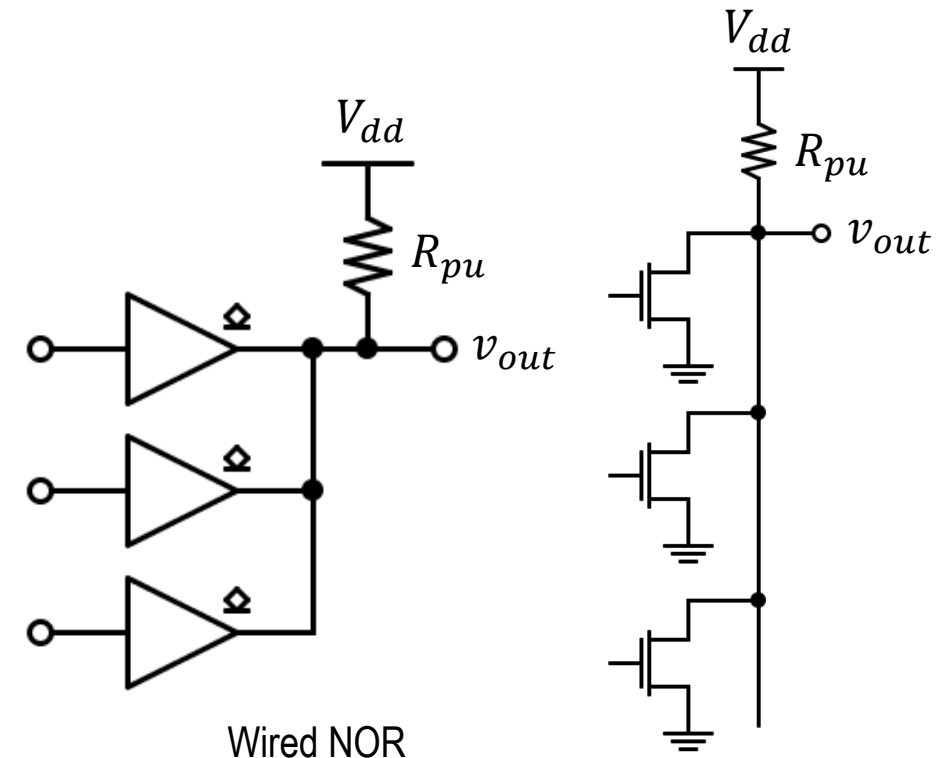
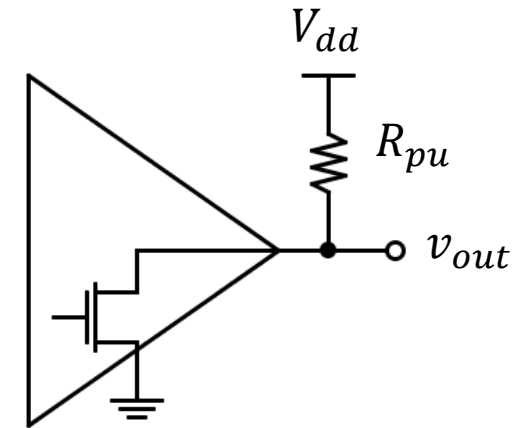
Two gates enabled
with opposite outputs
→ Short circuit



Single gate enabled
→ No conflict

Logic gates

- *Open collector or open drain output*
 - There is no pull-up transistor at the output
 - Can only set the output to logic 0
 - To set a logic 1
 - Switch off pull-down transistor
 - Use an external pull-up resistor
- Used to adapt logic levels with different voltage supplies
- Useful to drive the same line with multiple logic gates
 - No short circuit
 - When all logic gates set a logic 1
 - The output is at logic 1
 - When at least one logic gate sets a logic 0
 - The output is at logic 0
 - It is a NOR logic function



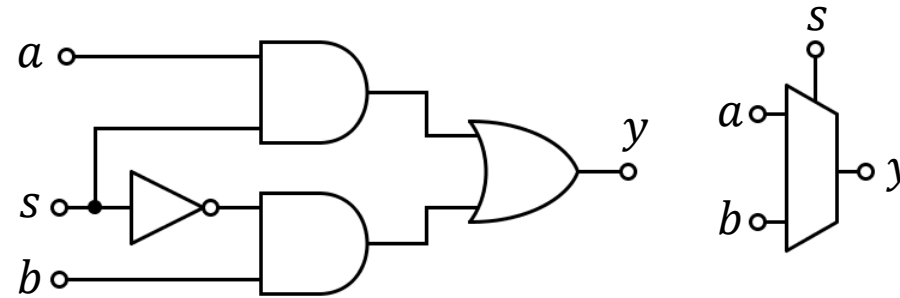
Combinational logic circuits

- The logic output depends on the current value of the inputs only
- Can be described using a *Truth Table*
 - List all possible input combinations and give the output values
- Synthesis of combinational logic circuits
 - Generate the circuit as a Sum of Products
 - Goals
 - Minimize the number of products
 - Minimize the number of literals in each product
 - Karnaugh maps
 - No more than 4 inputs in the exam
 - Quine-McCluskey algorithm for more inputs
 - Logic synthesis tools

Combinational logic circuits

- Multiplexer

- $y = a \cdot s + b \cdot \bar{s}$
- Selects either a or b as output



- Multiplexer on multiple bits

- Repeat the same structure multiple times

- Multiplexer with more than two inputs

- Need more than one selection signal

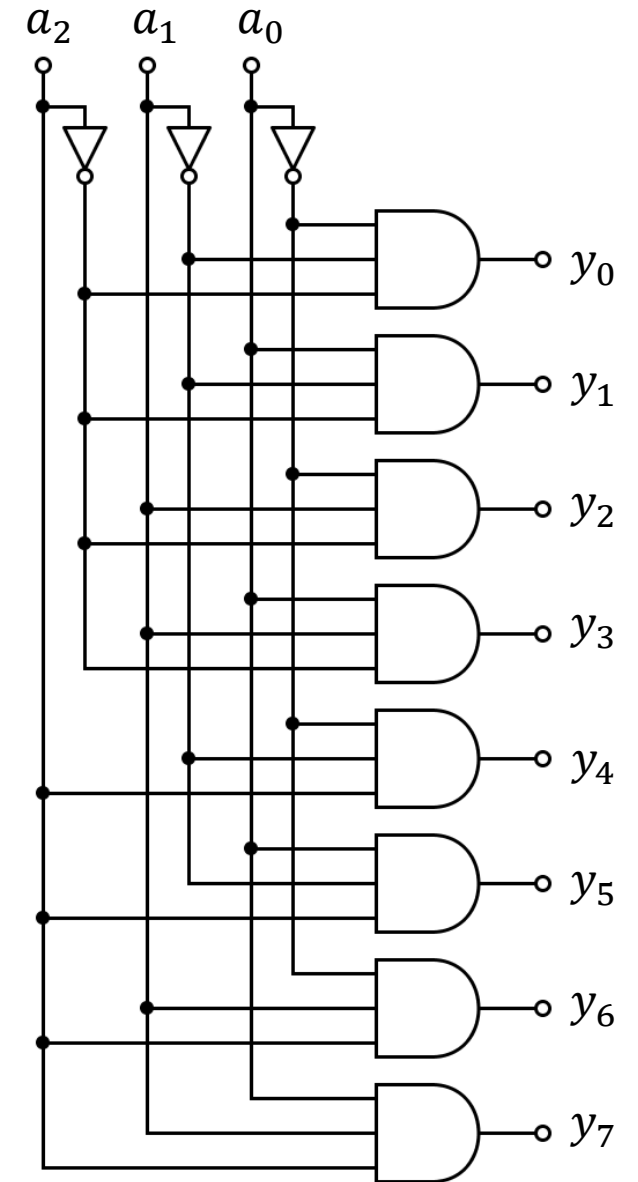
- Multiplexer with more than two inputs on multiple bits

- Combination of the above two cases

s	b	a	y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

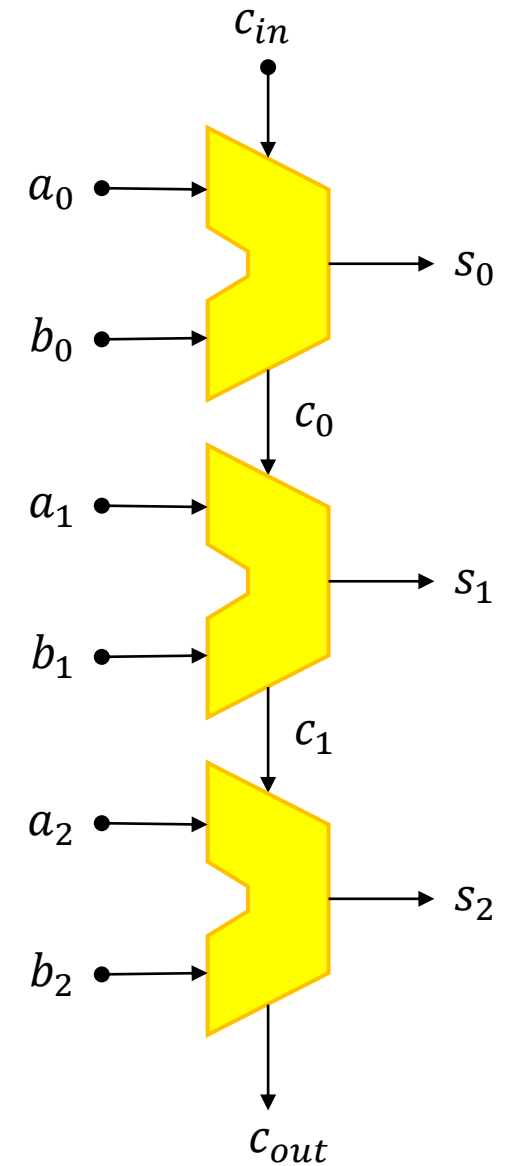
Combinational logic circuits

- Decoder
 - From n wires to 2^n wires
 - The n wires represent an integer in binary
 - A single wire of the 2^n outputs is active
 - Example: Address Decoder
- Encoder
 - From 2^n wires to n wires
 - A single wire of the 2^n inputs should be active
 - The n output wires represent the corresponding integer in binary
- More complex decoders/encoders
 - B4B5
 - Priority encoder
 - ...



Combinational logic circuits

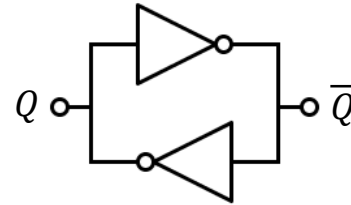
- Digital Adder/Subtractor
 - Half Adder and Full Adder
 - Operate on 1-bit operands
 - Generate a carry output
 - Multibit adders
 - Different area and performance based on carry computation
 - Ripple Carry
 - Simple but slow
 - Carry Look Ahead
 - Carry Select
 - Subtractor
 - Implemented using an adder (using 2-complement notation)
 - $A - B = A + \overline{B} + 1$



Sequential logic circuits

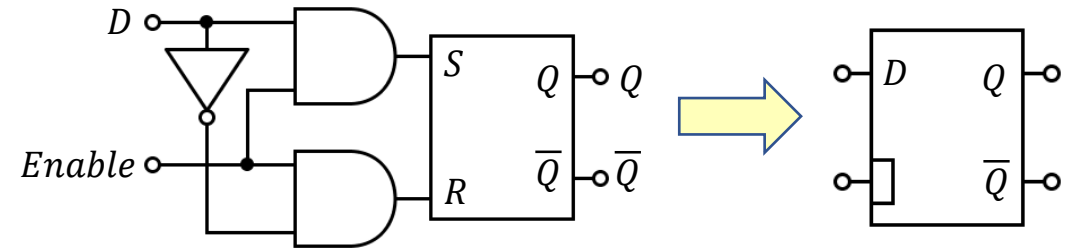
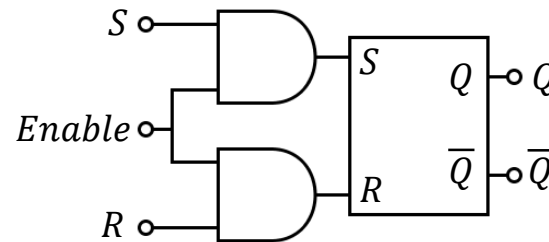
- Bistable circuit

- Two inverters in a loop
- SR Flip Flop (NOR or NAND)



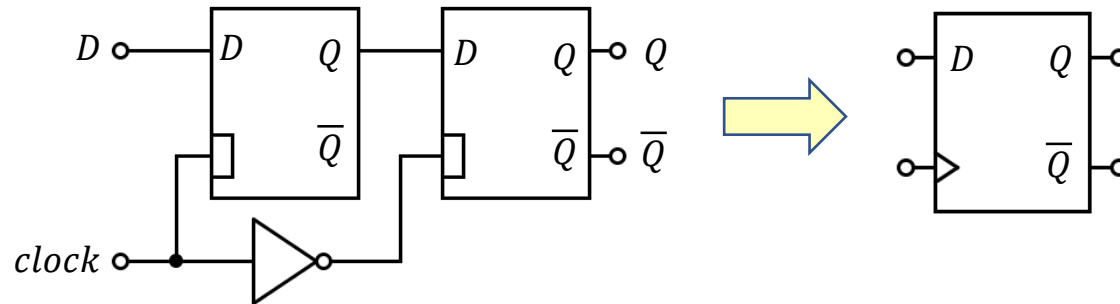
- SR FF with enable

- D-Latch
- Level sensitive



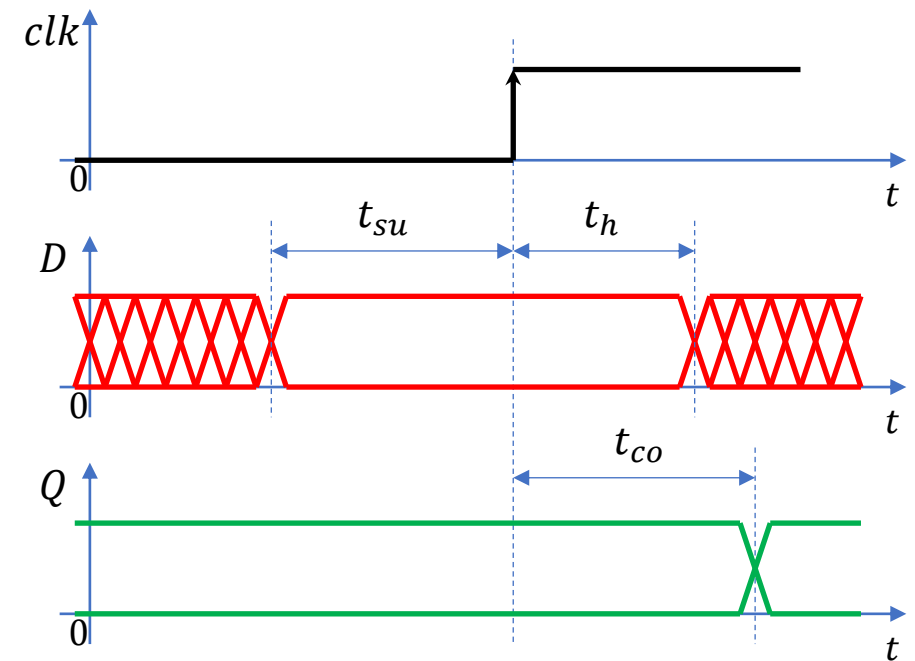
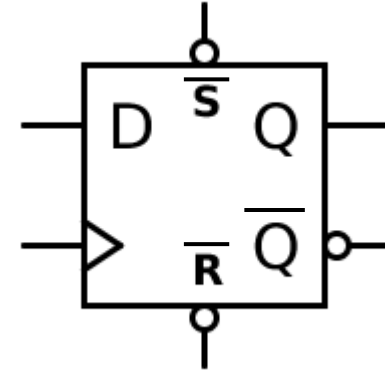
- Master-Slave D Flip Flop

- Two D-Latches
- Inverted enables
- Edge sensitive
- Clock signal



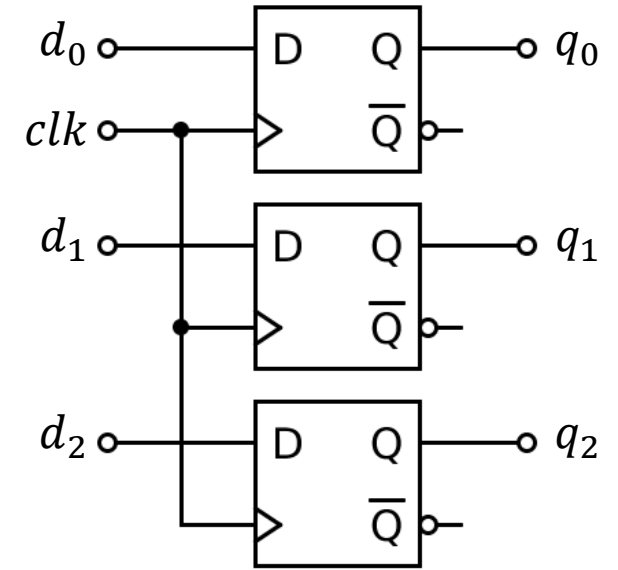
Sequential logic circuits

- Flip-Flop and Latches
 - Most devices have additional inputs
 - Asynchronous clear (or reset)
 - Sets the output to 0, without waiting the active clock edge
 - Asynchronous set
 - Sets the output to 1, without waiting the active clock edge
 - Clock enable
 - Masks the clock input
 - Signals are often active low
 - Timing specifications
 - Setup and Hold time: t_{su} and t_h
 - Clock to output: t_{co}

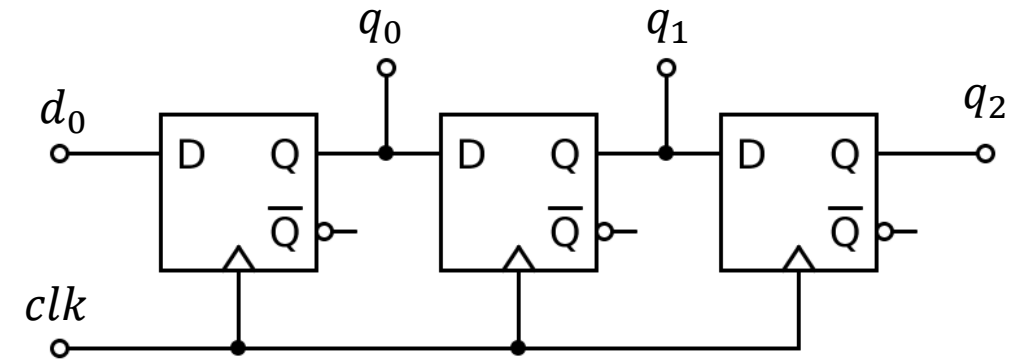


Sequential logic circuits

- Register
 - Combination of multiple Flip-Flops with a common clock
- Shift Register
 - Serial In, Parallel Out (SIPO)
 - Parallel In, Serial Out (PISO)
 - Additional control signal to load the parallel value in the register
 - Parallel Load (PL)
 - Load Enable (LE)
 - Serial In, Serial Out (SISO)
 - Often with Parallel Out, as well
 - All shift registers may have a shift enable signal



3-bit Register



3-bit SIPO Shift Register

Sequential logic circuits

- Counter

- Asynchronous counter

- Clock propagates from one FF to the next
 - Most significant bits are updated later than least significant bits

- Synchronous counter

- Same clock to all FFs
 - Internal logic used to compute the input to each FF
 - All output bits updated at the same time

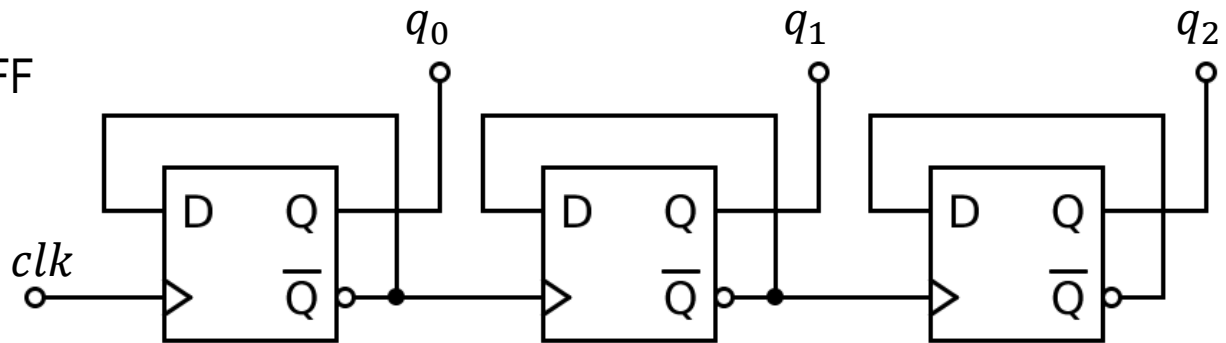
- Up or Up/Down counters

- Binary or Decade counters

- Parallel Load to initialize counter value

- Carry input and output to cascade counters

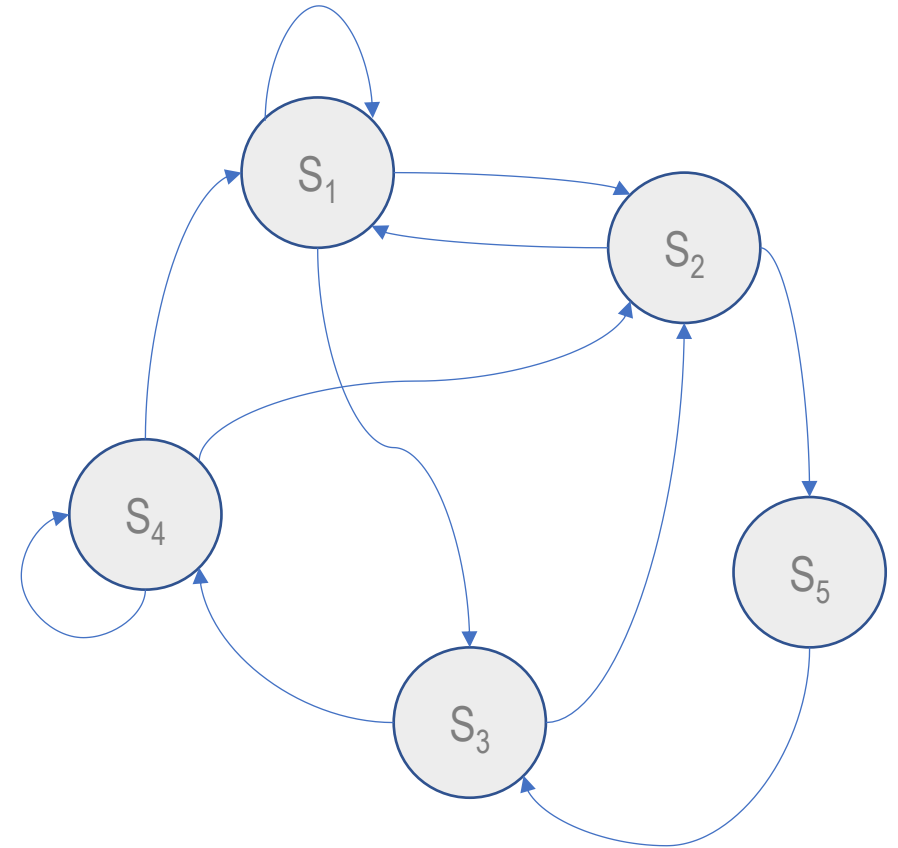
- Asynchronous reset



3-bit Asynchronous Counter

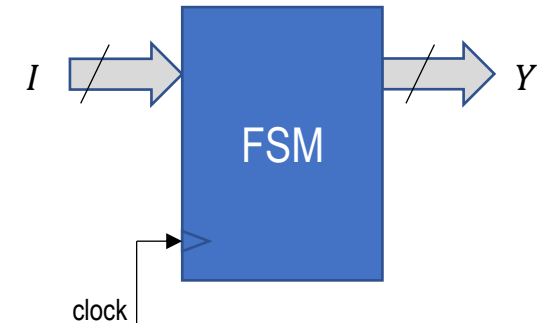
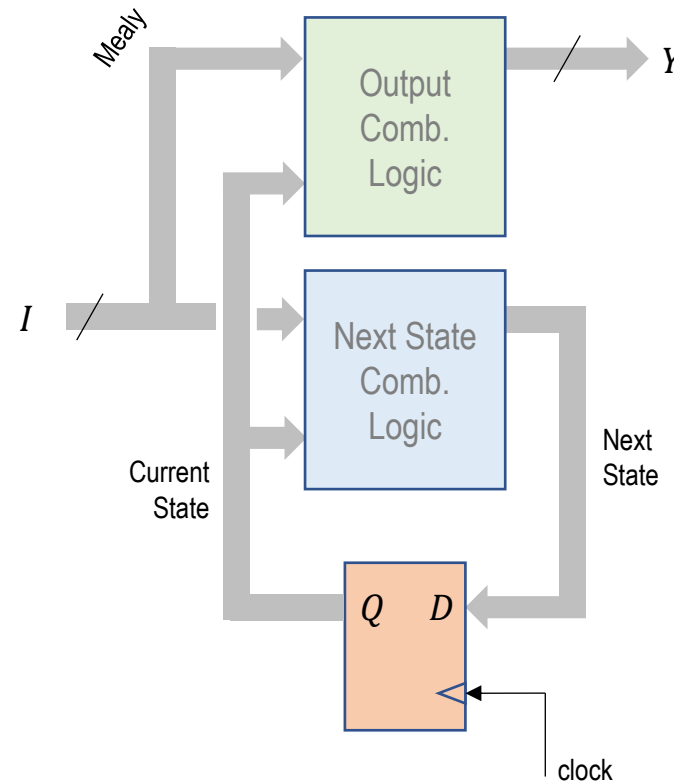
Sequential logic circuits

- Specification of an FSM
 - Use a state diagram
 - Edges have conditions on inputs
 - Edges may have output values (Mealy)
 - States may have output values (Moore)
 - Transitions are synchronized with an external clock signal
 - Write two truth tables
 - State transitions (next state)
 - With respect to inputs and current state
 - Outputs
 - With respect to current state (Moore) and inputs (Mealy)



Sequential logic circuits

- Specification of an FSM
 - The two truth tables become two combinational logic circuits
 - An n -bit register holds the current state
 - Binary encoding
 - Minimizes the number of bits
 - One-hot encoding
 - May simplify the combinational logic
 - Requires more bits in the register



VHDL

- Very high speed integrated circuit Hardware Description Language
 - Used to describe the structure and the behavior of a digital system
 - Used to *simulate* and *synthesize* a digital circuit
 - Syntax similar to a programming language
- Basic design element: ENTITY
 - It describes a block
 - The declaration describes the outside view of the block, by specifying the interface
 - Name of the entity
 - Ports
 - Inputs
 - Outputs
 - In/Out

```
entity fulladder is
  port (
    a, b, cin: in std_logic;
    s, cout: out std_logic
  );
end entity fulladder;
```

```
entity reg16 is
  port (
    D: in std_logic_vector(15 downto 0);
    clock, reset: in std_logic;
    Q: out std_logic_vector(15 downto 0)
  );
end entity reg16;
```

VHDL

- Basic design element: ARCHITECTURE
 - Describes the implementation of a block
 - Three ways
 - RTL (Register Transfer Level)
 - Uses concurrent signal assignments
 - Structural
 - Similar to a schematic diagram, but using a textual programming language
 - Behavioral
 - Uses a process, which is a sort of an endless loop

VHDL

- RTL
 - Concurrent statements
 - Boolean expression assigned to a signal
 - Executed in parallel, without any implied order
 - It is NOT a sequence of instructions
 - Can use high level operators
 - Arithmetic
 - Logic
 - You must include the proper libraries
 - Operators must be defined in the library for the data type
 - Can use conditional and selected signal assignments

```
c <= a AND b;  
d <= a OR c;  
s <= c + d;  
----- equivalent to:  
s <= c + d;  
c <= a AND b;  
d <= a OR c;
```

```
signame <= value1 when condition1 else  
           value2 when condition2;  
  
with expression select  
  signame <= value1 when choice1,  
                value2 when choice2,  
                value3 when others;
```

VHDL

- Structural
 - Declare components
 - Similar to an entity declaration
 - Similar to a function prototype in C
 - Declare signals
 - Instantiate components
 - port map
 - Assign signals to component ports
 - Can instantiate as many blocks as you want of the same component
 - Allows a hierarchical description of a circuit
 - Top/Down or Bottom/Up
 - Similar to designing a block diagram

```
entity ripple_carry is
  port (
    a, b: in std_logic_vector(7 downto 0);
    cin: in std_logic;
    s: out std_logic_vector(7 downto 0);
    cout: out std_logic
  );
end ripple_carry;
architecture rtl of ripple_carry is
  component fulladder is
    port (
      a, b, cin: in std_logic;
      s, cout: out std_logic
    );
  end component fulladder;
  signal c0, c1, c2, c3, c4, c5, c6: std_logic;
begin
  s0: fulladder port map (a(0), b(0), cin, s(0), c0);
  s1: fulladder port map (a(1), b(1), c0, s(1), c1);
  s2: fulladder port map (a(2), b(2), c1, s(2), c2);
  s3: fulladder port map (a(3), b(3), c2, s(3), c3);
  s4: fulladder port map (a(4), b(4), c3, s(4), c4);
  s5: fulladder port map (a(5), b(5), c4, s(5), c5);
  s6: fulladder port map (a(6), b(6), c5, s(6), c6);
  s7: fulladder port map (a(7), b(7), c6, s(7), cout);
end rtl;
```

VHDL

- Behavioral (process)

- In most cases uses an implicit wait statement on a sensitivity list
 - The process suspends when it reaches the end
 - Resumes when any signal on the sensitivity list changes
- Instructions in the process are sequential
 - But may be implemented in parallel to increase performance if the compiler finds it is feasible
- Can use *if* or *case* expressions
- Signal assignments happen only at the end
- Variables can be defined
 - Assignments to variables are immediate

```
process (reset, clock)
begin
  if (reset = '0') then
    cur_state <= init_state;
  elsif (rising_edge(clock)) then
    cur_state <= next_state;
  end if;
end;
```

```
if expression1 then
  seq_statement1;
elsif expression2 then
  seq_statement2;
else
  seq_statement3;
end if;
```

```
case expression is
  when choice1 =>
    seq_statement1;
  when choice2 =>
    seq_statement2;
  ...
end case;
```

VHDL

- FSM example

- Two processes

- One for the state register
 - One to compute next state and outputs

- Three processes (not shown)

- One for the state register
 - One to compute next state
 - One to compute the outputs

```
-- Entity declaration
entity fsm is (reset, clock)
  port (
    clk, reset: in std_logic;
    in1, in2: in std_logic;
    out1: out std_logic
  );
end entity fsm;
```

```
-- Architecture declaration
architecture behav of fsm is
  type FSMstatetype is (STA, STB, STC, STD);
  signal curstate, nextstate: FSMstatetype;
begin
  process (clk, reset)
  begin
    if (reset = '0') then
      curstate <= STA;
    elsif (rising_edge(clock)) then
      curstate <= nextstate;
    end if;
  end process;

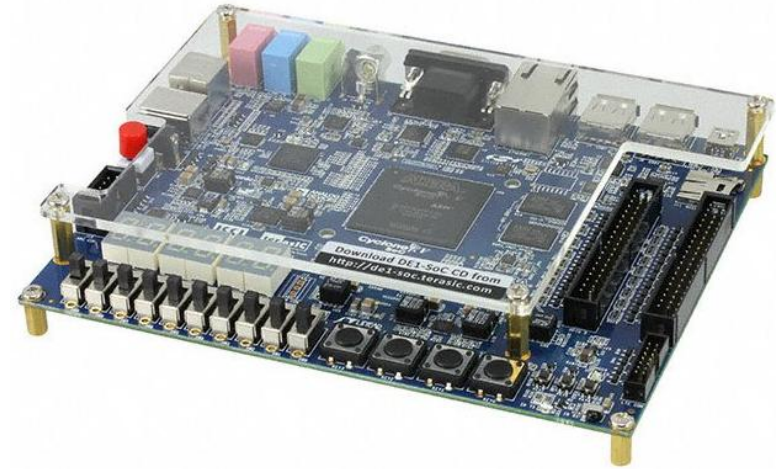
  process (curstate, in1, in2)
  begin
    nextstate <= curstate;
    case curstate is
      when STA =>
        -- Assign outputs
        -- Assign nextstate
      when STB =>
        -- Assign outputs
        -- Assign nextstate
      ...
    end case;
  end process;
end architecture;
```

Introduction to Embedded Systems

- This course and the entire degree program are focused on Embedded Systems
- What is an embedded system?
 - Electronic system
 - Hardware components
 - Software components
 - Communication systems
 - Part of a larger system, not necessarily electronic
 - Vehicles (cars, planes, ...)
 - Industrial systems
 - Home appliances
 - ...
 - Examples of Embedded Systems

Introduction to Embedded Systems

- How does an embedded system look like?
 - It is an electronic system with:
 - Printed Circuit Board (PCB)
 - Digital components
 - Processors, memories, I/O, ...
 - Analog components
 - Power management, amplifiers, AD and DA converters, ...
 - Interface components
 - Sensors and actuators
 - Software components
 - It looks like a PC
 - But it is not perceived as a PC
 - Sometimes it is not perceived at all



Introduction to Embedded Systems

- What are the characteristics of an embedded system?
 - It is embedded, i.e. it lives in a larger environment
 - It should **react**, typically in real-time, to external stimuli
 - It is dedicated to a particular application (single task)
 - Possibility to better optimize
 - Actually forced to optimize as much as you can
 - Tightly constrained with respect to several metrics
 - Performance, costs, size, power consumption, etc.
 - Very competitive market
 - Very large market
 - Global embedded system market worth \$214 billion by 2020
 - It was \$140 billion in 2013 (compound annual growth rate 6.5%)
 - Driven by automotive, healthcare, Internet of Things, ...

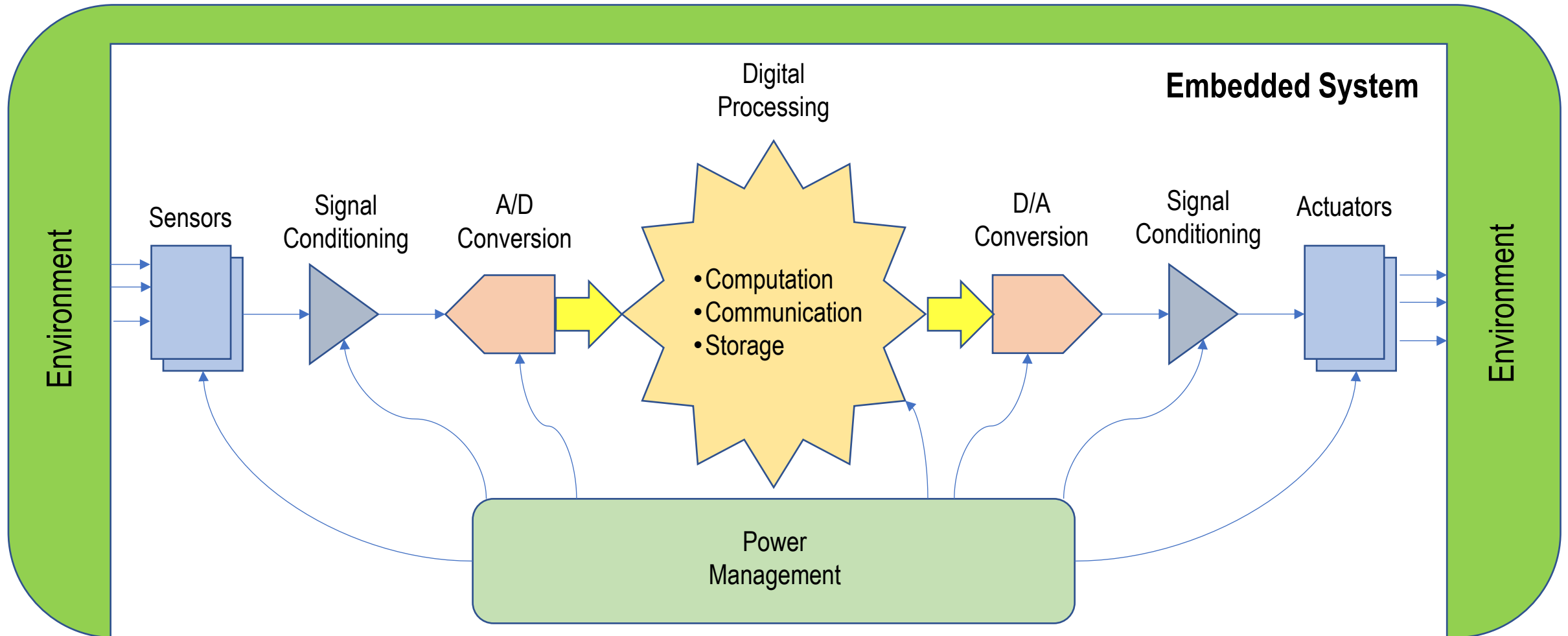
Introduction to Embedded Systems

- Is a PC an embedded system?
 - No, it has the same internal structure, but it is not an embedded system
 - It is **interactive** with the user
 - An embedded system is reactive
 - It should perform several different tasks
 - Optimizations are trade-offs between very different requirements
- Your goal after graduation
 - Design embedded system
 - Difficult because the design space is extremely large
 - Difficult because it requires skills from many very different domains
 - Simultaneously optimize numerous design metrics



Introduction to Embedded Systems

- Typical architecture of an Embedded System



Introduction to Embedded Systems

- Design metrics
 - A measurable feature of a system implementation
 - Common metrics
 - Unit cost
 - Non-Recurring Engineering (NRE) cost
 - Size
 - Performance
 - Power
 - Flexibility
 - Time-to-Market
 - Safety
 - ...

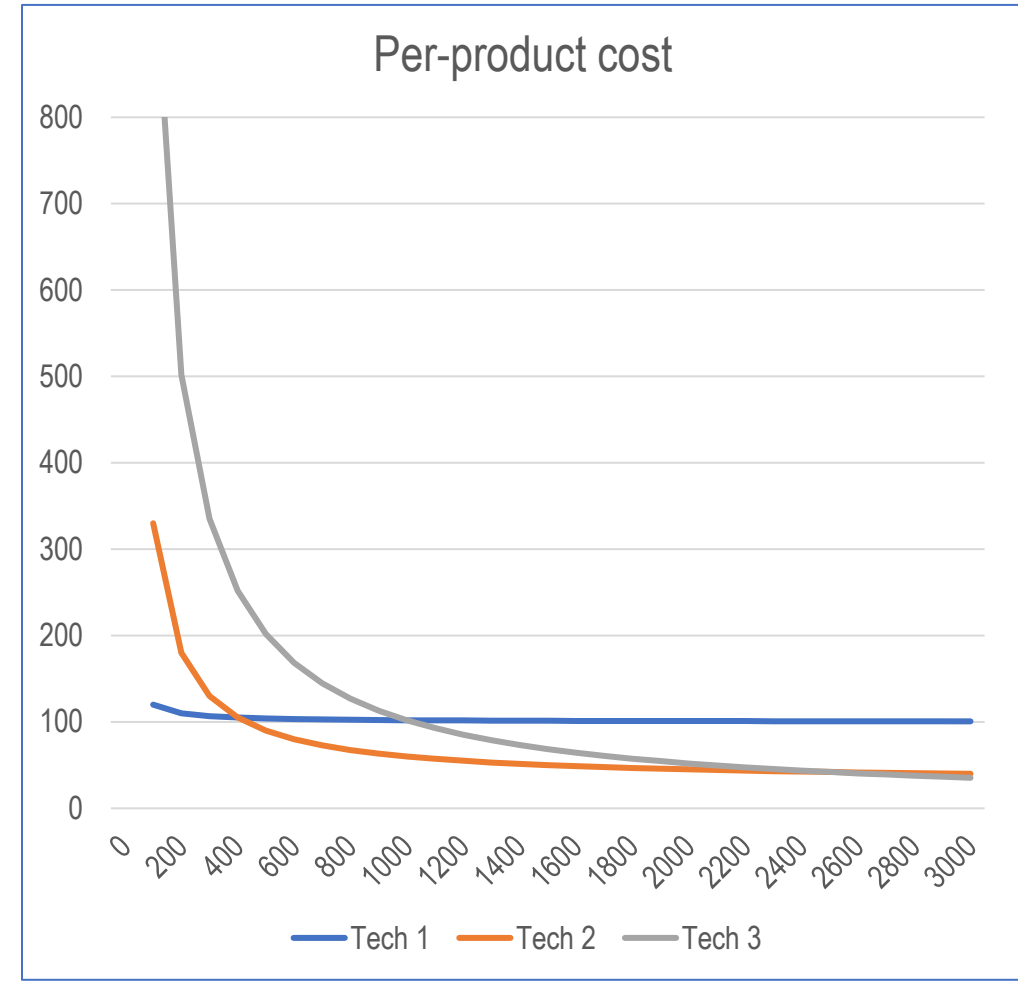
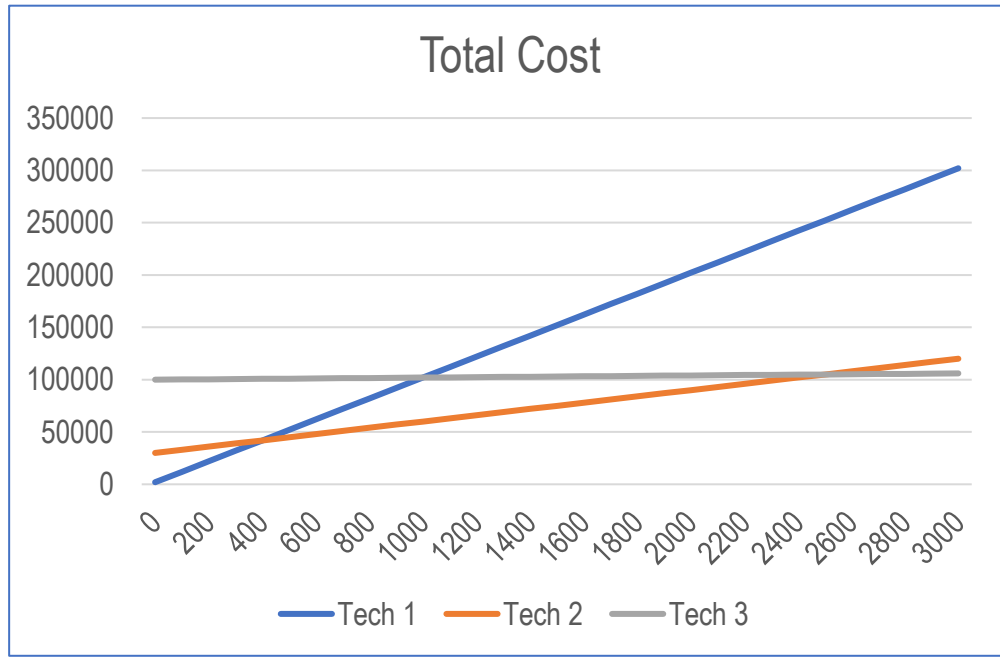
Introduction to Embedded Systems

- Unit cost U
 - Monetary cost of manufacturing each copy of the system
- NRE cost N
 - One-time monetary cost of developing (designing) the system
- Total cost T
 - $T = N + U \cdot n_{unit}$
- Per-product cost p
 - $p = \frac{T}{n_{unit}} = \frac{N}{n_{unit}} + U$
- Different technologies have different U and N

Introduction to Embedded Systems

- Example

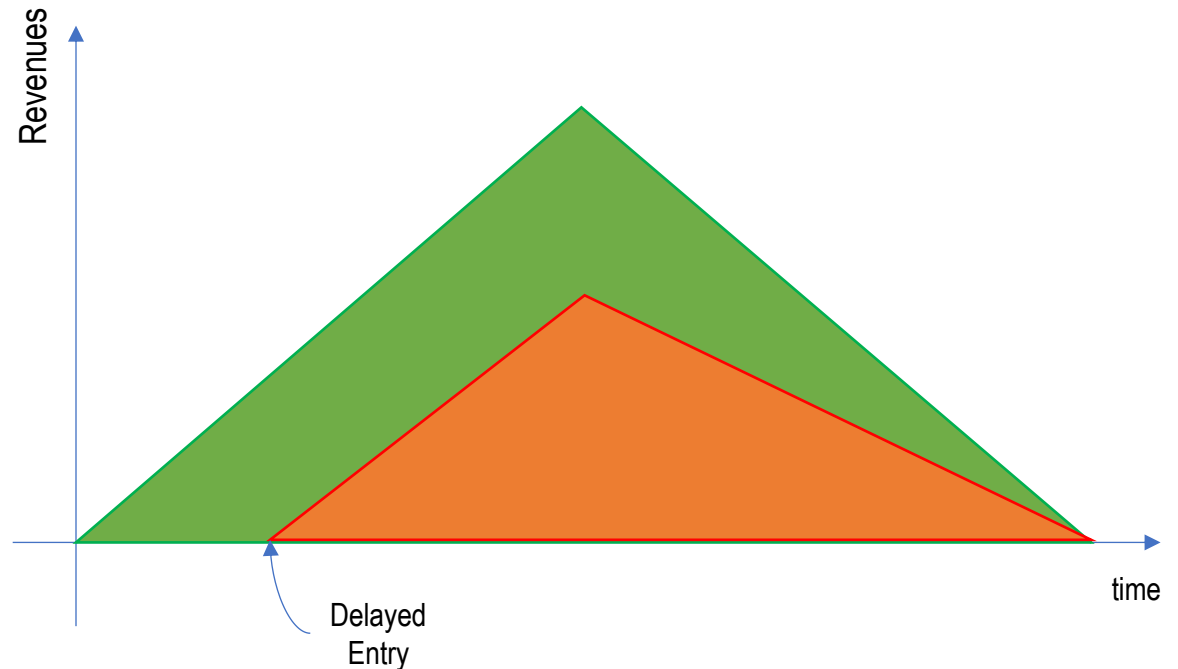
- Tech1: NRE = \$2000, Unit = \$100
- Tech2: NRE = \$30000, Unit = \$30
- Tech3: NRE = \$100000, Unit = \$2



Introduction to Embedded Systems

- Time-to-Market

- Time required to develop a product to the point it can be sold to customers
- Losses due to delayed market entry
 - Total revenue is the area of the graph
 - Loss is the difference in area
- Different technologies give different Time-to-Market
 - Choose a technology to enter the market
 - Later refine the product with a better technology



Introduction to Embedded Systems

- Performance
 - Can be expressed in several ways
 - Latency (response time)
 - Time between task start and end
 - Throughput
 - Tasks per second
 - Parallelism increases performance
 - No parallelism
 - $Thr = 1/Lat$
 - n parallel blocks
 - $Thr = n/Lat$

Introduction to Embedded Systems

- Technology
 - A way of accomplishing a task
 - Technical processes
 - Methods
 - Knowledge
- Technologies for embedded systems
 - Processor technology
 - IC technology
 - Design technology
- Technologies are orthogonal

Introduction to Embedded Systems

- Processor technology
 - General Purpose (GP) Processor
 - It's the classic programmable micro-processor you are used to
 - Embedded micro-processors are different from PC micro-processors
 - They are within a System-On-Chip (SoC)
 - They include a number of peripherals
 - Designed to meet several different requirements
 - Not optimized for any specific application
 - Cheap
 - Billions of units sold every year
 - Flexible, low time-to-market
 - Software is easy to update and modify
 - No high performance
 - Compared to a dedicated design
 - Not power efficient

Introduction to Embedded Systems

- Processor technology
 - Application Specific Instruction Processor (ASIP)
 - A general purpose processor with specific hardware for a certain application domain
 - Example: DSP
 - Dedicated to signal processing application
 - Custom instruction: Multiply and Accumulate (MAC)
 - Example: micro-controller
 - Dedicated hardware to drive and read external pins
 - Cheap
 - Flexible, low time-to-market
 - Higher performance and better power than a GP processor

Introduction to Embedded Systems

- Processor technology
 - Single Purpose (custom) Processor
 - Digital hardware designed specifically for a certain application
 - Advantages
 - Best performance
 - Best power consumption
 - Small size
 - Disadvantages
 - Long time-to-market
 - Very high NRE costs
 - Low flexibility
 - Good for large volume productions
 - Good for extremely high performance designs

Introduction to Embedded Systems

- IC technology
 - Full Custom (VLSI)
 - The integrated circuit is entirely designed and optimized for a particular application
 - Transistor sizing, placing and routing
 - Advantages
 - Best for performance, size and power consumption
 - Disadvantages
 - High NRE costs
 - All masks in the production process are dedicated to a single design
 - Long time-to-market
 - Currently used to design
 - Micro-processor (GPs and ASIPs)
 - Memories
 - Programmable Logic

Introduction to Embedded Systems

- IC technology
 - Semi-Custom (Gate Array / Standard cells)
 - The lower layers are fully or partially built
 - Placing of large blocks, routing
 - Advantages
 - Good performance, size and power consumption
 - Lower NRE compared to a full custom implementation
 - Some masks are shared among more than one design
 - Disadvantages
 - High NRE costs
 - Long time-to-market

Introduction to Embedded Systems

- IC technology
 - Programmable Logic Device (PLD)
 - The integrated circuit is fully built by the manufacturer
 - The embedded system designer buys the IC
 - The PLD can be customized after fabrication for a specific application
 - Logic functions and interconnections are programmable
 - Advantages
 - Low NRE costs
 - Fairly good performance and power consumption
 - But worse than full custom and semi-custom
 - Disadvantages
 - High unit cost
 - Large size

Introduction to Embedded Systems

- Processor and IC technologies are orthogonal
 - The design space is very large
 - Very long to explore
 - Early design decisions are difficult
 - Use modern design technologies to evaluate different solutions

